BY ORDER OF THE COMMANDER
RAMSTEIN AIR BASE (USAFE)

DEPARTMENT OF THE AIRFORCE
MANUAL

13-204

RAMSTEIN AIR BASE
Supplement

25 APRIL 2024

Space, Missile, Command and Control

AIRFIELD OPERATIONS

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: This publication is available for downloading or ordering on the e-Publishing website at **www.e-Publishing.af.mil**.

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: 86 OSS/OSA

Certified by: 86 OSS/CC Lt Col Taylor S. Rigollet

Supersedes: RAMSTEIN AIR BASE INSTRUCTION 13-204, 15 February 2019

Pages: 80

This publication implements Air Force Policy Directive (AFPD) 13-2, Air Traffic, Airfield, Airspace and Range Management. It describes the operational environment for Ramstein Air Base (AB) and the local area and prescribes policies/procedures for air traffic control (ATC), airfield management (AM), airspace, and flight line activities. The provisions of this instruction are directive in nature and apply to all assigned/attached units and Air Force Reserve and Air National Guard personnel. Pilots may deviate from the procedures contained herein in the interest of flight safety or when directed by Langen Air Control Center (ACC), Ramstein Ground Controlled Approach (GCA), Ramstein Tower, or other competent authority. Note: All altitudes in this instruction are mean sea level (MSL) unless specified otherwise or when referring to weather minimums. Aircrew and ground personnel should refer to Notices to Airmen (NOTAM) and AM for up-to-date information. This publication may be supplemented at any level, but all direct supplements must be routed to the Office of Primary Responsibility (OPR) of this publication for coordination prior to certification and approval. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the AF Form 847, Recommendation for Change of Publication; route DAF Forms 847 from the field through the appropriate functional chain of command. Ensure that all records created as a result of processes prescribed in this publication are maintained IAW Department Air Force Instruction (DAFI) 33-322, Records Management and Information Governance Program, and disposed of IAW Air Force

Records Information Management System (AFRIMS) Records Disposition Schedule (RDS). The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF **CHANGES** This document has been substantially revised and must be completely reviewed. Major changes have been made throughout the entire publication.

Chapter 1

GENERAL INFORMATION

1.1. Roles and Responsibilities.

1.1.1. The 86 Operations Group Commander (86 OG/CC) is responsible for flying operations at Ramstein AB.

1.1.2. The Station 1 (Building 2303) Fire Chief (86 CES/CEF) or designated rep (Chief 2) is the initial on-scene commander for ground emergencies and in-flight emergencies IAW **Paragraph 5.2.1** of this instruction.

1.1.3. The Control Tower provides air traffic services to aircraft operating within the Ramstein Class Delta Airspace. Ramstein Ground Controlled Approach (GCA) provides air traffic services to aircraft operating within the lateral boundaries described in **Attachment 7**.

1.1.4. The 786th Civil Engineering Squadron (786 CES) is responsible for the operation and maintenance of the aircraft arresting systems. Power Production (786 CES/CEOFP) provides indoctrination training briefing of arresting systems to ATC and AM personnel.

1.1.5. The 86th Maintenance Squadron (86 MXS) is responsible for the training and response of crash recovery personnel.

1.1.6. Wing Safety (86 AW/SE) manages the Bird-Aircraft Strike Hazard (BASH) program.

1.1.7. The 86th Operations Support Squadron Commander (86 OSS/CC) and 86th Operations Support Squadron Director of Operations (86 OSS/DO) are responsible for approving, denying, and coordinating non-standard training, exercises, and operations on Ramstein AB ramps, taxiways, and runways. This includes use of Ramstein Drop Zone (DZ), Landing Zone Safety Officer (LZSO) training, construction, military ceremonies, and use of ramps by an organization other than the controlling agency listed in **Paragraph 1.10**.

1.1.8. All assigned and Temporary Duty (TDY) units are responsible for safe air and ground operations on the airfield. Use of the airfield for events beyond day-to-day operations must be coordinated through the 86th Operations Support Squadron 86 OSS; otherwise, Airfield Management (AM) has the authority to stop the event in the interest of safety.

1.2. Airfield Description: Ramstein Air Base (ICAO: ETAR) is in the southwest portion of Germany, in the federal state of Rheinland-Pfalz, in the rural district of Kaiserslautern, adjacent to the small town of Ramstein-Miesenbach. The airfield is situated on the low-lying south side of the installation. See **Attachment 2** for airfield layout.

1.2.1. Ramstein AB aerodrome reference point is 49˚26.17’ N and 7˚36.39’ E.

1.2.2. **The airfield elevation is 781 feet.**

1.2.3. Ramstein AB has two runways designated as Runway 08/26 and Runway 09/27.

1.2.3.1. Runway 08/26 is considered the runway reference point and is located at 49˚26.04’ N and 7˚36.02’ E.

1.2.3.1.1. Runway 08/26 is 10,497 feet long and 148 feet wide; the entire usable length is grooved.

1.2.3.1.2. The first 1,818 feet of Runway 08 and the first 1,836 feet of Runway 26 are Portland cement concrete (PCC). The remaining 6,843 feet is asphalt concrete (AC) or porous friction coarse asphalt.

1.2.3.1.3. Both ends of Runway 08/26 have 665-foot AC overruns.

1.2.3.2. Runway 09/27 is 9,841 feet long and 148 feet wide; the entire usable length is grooved.

1.2.3.2.1. The first 1,401 feet of Runway 09 and the first 1,410 feet of Runway 27 is PCC. The remaining 7,030 feet is AC.

1.2.3.2.2. Both ends of Runway 09/27 have 665-foot AC overruns.

1.2.3.3. Runway gradients:

1.2.3.3.1. Runway 08 gradient is +0.1429%.

1.2.3.3.2. Runway 09 gradient is +0.1930%.

1.2.3.3.3. Runway 26 gradient is -0.1429%.

1.2.3.3.4. Runway 27 gradient is -0.1930%.

1.2.4. Runway VFR/Instrument hold line locations. Runway VFR hold lines are located at least 352 feet from the centerline (277 feet from the edge) of all runways IAW International Civil Aviation Organization (ICAO) Annex 14 and are located on Taxiways Alpha, Bravo, Charlie, Echo, Golf, Kilo, and Lima. Instrument hold lines are placed to protect the glideslope and localizer critical areas and are located on Taxiways Alpha, Charlie, Echo South, Golf (North and Middle), Lima and Runway 09/27 near Taxiway Echo. See **Attachment 2** for runway/instrument hold line locations.

1.2.5. All taxiways on Ramstein are 75 feet wide except for Taxiway Golf (South), which is 49 feet wide.

1.3. Runway Selection Procedures.

1.3.1. Runway 26 is the designated primary calm wind runway. Runway 26 shall be the runway in use whenever the prevailing wind speed is less than 5 knots regardless of wind direction.

1.3.2. Runway 08/26 is the designated Primary Instrument Runway.

1.3.3. The tower watch supervisor may designate a runway in use other than that favored by the prevailing wind when an operational advantage will be gained.

1.3.4. Pilot requests for opposite direction operations will be approved on a workload- permitting basis.

1.3.5. Runway change notification.

1.3.5.1. Tower will verbally coordinate with GCA prior to the runway change and notify AM and Weather after a runway change.

1.3.5.2. AM will notify the Fire Department alarm room, Barrier Maintenance (786 CES/CEOFP), Command Post (86 AW/CP), Transient Alert (TA), Maintenance Operations Centers (MOC) (86 MXG/MXOC and 721 AMXS/MOC), Air Mobility Control Center (721 MSS/AMCC) after a runway change.

1.3.6. Simultaneous runway operations are not authorized at Ramstein AB.

1.4. Controlled Movement Area (CMA).

1.4.1. The northern boundary of the CMA starting at the northwest corner follows 10 feet inside of the edge of West Perimeter Road going east. Defined by the perimeter road north of Bldg. 4456, encompassing the road, the CMA extends south to run parallel with the Taxiway Golf North VFR hold line to Taxiway Echo North VFR hold line. The CMA then extends in a straight line from the Taxiway Echo North VFR hold line to the Runway 27 glideslope access road instrument hold line. Continuing east of the glideslope access road instrument hold line, the upper boundary of the CMA follows the glideslope access road, encompassing the road, so that a vehicle crossing Taxiway Charlie or turning west or east off Taxiway Charlie onto the glideslope access road is required to request CMA access. Vehicles traveling south on Taxiway Charlie must request CMA access prior to the Taxiway Charlie VFR hold line.

1.4.2. The southern boundary is defined by the Morbach drainage ditch, located approximately 700 feet south of the centerline of Runway 08/26. Exception: The CMA is marked by the VFR hold lines on Taxiways Echo South, Lima and Golf South, excluding buildings 2398 and 2399.

1.4.3. The eastern boundary of the CMA follows 10 feet inside of the edge of the east perimeter road from the North Crash Access Road to a point parallel with the Morbach drainage ditch south of Runway 08/26.

1.4.4. The western boundary follows 10 feet inside the edge of the west perimeter road from Taxiway Golf (North) to Taxiway Golf (South).

1.4.5. Vehicles/pedestrians operating in the CMA. A detailed outline of Ramstein AB flight line vehicle/pedestrian operations can be found in local airfield driving directives (AFI-13-213_RAMSTEINSUP, Airfield Driving). Only mission essential government vehicles and other approved vehicles displaying the appropriate flight line pass are permitted in the CMA. No driver will operate a vehicle on any portion of the airfield in MOPP4 gear unless a real-world emergency exists. Vehicles operating inside the CMA are required to remain outside of 125 feet from the edge of the runway, unless given approval onto the runway from tower.

1.4.5.1. Due to wingtip clearances, aircraft with a wingspan greater than 142 feet are not permitted to cross the landing threshold or begin departure roll while any vehicle/pedestrian is operating within 125 feet of the runway.

1.4.5.2. Emergency Vehicles.

1.4.5.2.1. Any emergency vehicle requiring access to the CMA will obtain permission for entry from tower. Fire Department vehicles will utilize the Crash Net to communicate with tower. All other vehicles will use the Ramp Net.

1.4.5.2.2. To maintain response reaction times, limited fire response vehicles (due to their size and maneuverability) may utilize Taxiway Golf to access Fire Station #3 in the southwest area.

1.4.6. Gaining Access to the CMA. Aircraft, vehicles, or personnel will not operate in the CMA without two-way radio communication and approval from tower. Approval for entry to CMA via phone calls to tower or AM by third parties is prohibited, including during exercises or emergencies. If circumstances beyond an individual's control preclude radio communication, they must obtain an escort with two-way radio capability before being granted access to the CMA.

1.4.7. Exiting the CMA. All vehicles and personnel must advise the tower when exiting the CMA. This is normally done via the Ramp Net or Crash Net.

1.4.8. Air traffic controllers use a light gun as a backup system for communicating with aircraft or ground vehicles if their radios fail. When a vehicle operator experiences a radio failure on a runway or taxiway, they must vacate the runway as quickly and safely as possible and contact the tower or AM by other means, such as a cell phone, to advise of the situation. If this is not practical, then the driver, after vacating the runway, should turn the vehicle toward the tower and start flashing the vehicle headlights and wait for the controller to signal with the light gun IAW Air Force Visual Aid 11-240.

1.5. Mitchell Avenue Crossing.

1.5.1. The activation of traffic lights is an additional service and will not take precedence over the control and movement of aircraft. Vehicles traveling on Mitchell Avenue will give way to aircraft and vehicles utilizing Taxiway Delta at all times. Workload permitting, tower will activate the traffic light:

1.5.1.1. Anytime an aircraft is taxiing on Taxiway Delta and will cross Mitchell Avenue, the light shall remain on from approximately 2 parking spots prior and after the intersection.

1.5.1.2. When aircraft are taxiing into Ramp 3 Spot 1 or Ramp 2 Spot 12, the light shall remain on from approximately 2 parking spots prior until the aircraft is established in parking.

1.5.2. AM is responsible for inspecting/tracking all maintenance actions associated with vehicle traffic lights/warning sirens on the airfield.

1.6. Runway Crossing for Vehicles. All vehicles requesting to cross Runway 08/26 or Runway 09/27 at Taxiway Echo or Taxiway Golf will state which portion they are requesting to cross from and where they will terminate the crossing. Portions requiring clarification are Taxiway Echo North, Echo Middle, Echo South, Golf North, Golf Middle and Golf South. Example: “Airfield 3, request to cross Runway 26 from Taxiway Echo North to Taxiway Echo Middle.” Multiple runway crossings are not authorized at Ramstein AB.

1.7. Airfield Lighting Systems. Tower is equipped with the capability to operate airfield lighting systems and visual aids. Tower will operate the airfield lighting system IAW FAAO JO 7110.65, Zentralrichtlinie Militärisches Flugverkehrsmanagement A2-272/2-2000-14 (German ATC publication) and the following local procedures, or pilots’ requests.

1.7.1. Lighting system types.

1.7.1.1. Runway lights.

1.7.1.1.1. Runway 08: High intensity runway lights (HIRL) with 5-step intensity, runway centerline lights (RCL), and runway end identifier lights (REIL).

1.7.1.1.2. Runway 26: HIRLs, RCLs, REILs, and touchdown zone lights.

1.7.1.1.3. Runway 09/27: HIRLs, REILs.

1.7.1.2. Approach lights.

1.7.1.2.1. Runways 08, 09, 27: Approach Light System with Sequenced Flashing Lights (ALSF-1) shortened to NATO standard Approach Light System with Sequenced Flashing Lights (SFL) 5-step intensity.

1.7.1.2.2. Runway 26: ALSF-2 and SFL.

1.7.1.3. Precision approach path indicator (PAPI): Runways 08/26 and 09/27, 5-step intensity.

1.7.1.4. Taxiway lights: All taxiways are lighted with standard blue taxiway lights, 1-step intensity. Taxiways Alpha, Bravo, Echo, Foxtrot, Golf (north of Runway 08/26); Kilo, Lima, and Runways 08/26 and 09/27 have taxiway centerline lighting. Exception: The taxitracks south of Ramp 7 are unlit.

1.7.1.5. Stop bar lights: Located at hold lines on Taxiways Alpha (east of Taxiway Bravo), Bravo (south of Runway 09/27), Charlie, Echo (north of Runway 09/27, north of Runway 08/26 and south of Runway 08/26), Golf (north and south of Runway 08/26, and north of Runway 09/27) Kilo, and Lima; and on Runway 09/27 (east and west of Taxiway Echo). IAW Zentralrichtlinie Militärisches Flugverkehrsmanagement A2-272/2-2000-14, aircraft or vehicles shall not cross an illuminated stop bar without prior approval from ATC.

1.7.1.6. Threshold lights: Runways 08/26 and 09/27.

1.7.1.7. Distance remaining markers: Runways 08/26 and 09/27. Note: Distance Remaining Markers on Runway 08/26 are located 92 feet from the runway edge.

1.7.1.8. Rotating beacon.

1.7.1.9. Runway Guard Lights. The “Wig Wag” lights are to provide a distinctive warning to anyone approaching the runway holding position that they are about to enter an active runway. The lights are installed abeam the VFR hold lines at the intersections of Runway 09/27 at Taxiways Charlie, Echo North, and Golf North. Additionally, there are lights located at the intersections of Runway 08/26 at Taxiways Golf South, Echo South, and Lima.

1.7.2. Lighting System Operations.

1.7.2.1. Unless requested, airfield lights will be turned off between sunset and sunrise when not needed for air or ground aircraft movement operations to conserve energy.

1.7.2.2. Between sunset and sunrise, tower will activate all necessary lighting when taxi instructions are issued to an aircraft. Tower will leave these lights on until aircraft departure or shutdown.

1.7.2.3. Airfield lights may be adjusted to Step 5 to aid in snow removal operations, if doing so does not negatively affect flight operations.

1.7.3. Lighting System Outages. AM is the POC for airfield lighting outages, except for the lighting panel.

1.7.3.1. AM will notify Airfield Lighting (duty hours) or the CE service desk (after duty hours) of outage and request estimated time of fix actions. Customer Service will then contact the appropriate section(s) to respond to the emergency call. Responding member has a 1-hour response time from the time they have received the call.

1.8. Closed/Unusable Portions of the Airfield. Ramp 6, Taxiway Hotel, and access taxitracks into hardened aircraft shelters on Taxitrack Golf South are closed until further notice.

1.9. Aircraft Arresting Systems.

1.9.1. Ramstein AB is equipped with Barrier Arresting Kit (BAK)-12/14 systems on Runway 08/26. See **Attachment 2**, Airfield Diagram and CMA for depiction of locations. Note: There are no arresting systems on Runway 09/27.

1.9.1.1. The west BAK-12/14 is 1,815 feet from Runway 08 threshold.

1.9.1.2. The east BAK-12/14 is 1,833 feet from Runway 26 threshold.

1.9.2. Configuration. Barriers will be connected and in the down position. The departure end barrier will be raised for all tail hook-equipped aircraft departing/arriving and upon request.

1.9.3. Responsibilities:

1.9.3.1. 786 CES is responsible for the operation and maintenance of the aircraft arresting systems.

1.9.3.2. 86 MXS is responsible for the training and response of crash recovery personnel.

1.9.3.3. Tower controllers will report all arresting system malfunctions to AM. Tower will notify AM before releasing arresting systems to barrier maintenance for maintenance or configuration changes.

1.9.3.4. AM will notify 786 CES/CEOFP and the tower of all arresting system malfunctions or abnormalities.

1.9.3.5. Barrier maintenance will notify AM before conducting maintenance which requires the use of tools on the arresting systems.

1.9.4. Barrier engagement. See **Attachment 3**, Aircraft Arresting Barrier Engagement Procedures, for information regarding barrier engagement, responsibilities, suspension, and resumption of runway operations, etc.

1.10. Parking Plan and Restrictions.

1.10.1. AM manages seven ramps on the aerodrome and remains the overall authority for ensuring proper airfield utilization.

1.10.1.1. Ramp 1 is operated by the 86th Aircraft Maintenance Squadron (86 AMXS) and has 20 aircraft parking spots available to accommodate base-assigned C-130s. Aircraft on Spots 1-1 to 1-11 and 1-13 to 1-18 will face west. Aircraft on Ramp 1 hot spots, located on the south side of the ramp, will face south. Spot 1-12 does not exist. See **Attachment 4**, Airfield Parking Ramps 1, 2, 3, AND 4.

1.10.1.2. Ramp 2 is operated by 86 AMXS and has 12 aircraft parking spots available to accommodate base-assigned C-130s and C-37s. Aircraft on Spots 2-1 to 2-12 will park facing north if the aircraft has the capability to reverse taxi. Base assigned C-37s will face west on Spots 2-1 and 2-2. Parking spots oriented east/west are designed to accommodate aircraft with a wingspan of 130 feet or less; if parking spots are utilized to accommodate C-130 parking, prior coordination with AM is required to ensure appropriate wingtip restrictions are imposed for Taxiway Delta. Spots 2-1 through 2-8 will be used for normal operations. Spots 2-9 through 2-12 will only be utilized when use of other available parking spots will impede operations. Coordination with AM is required for use of Spots 2-9 to 2-12. See **Attachment 4**, Airfield Parking Ramps 1, 2, 3, 4.

1.10.1.3. Ramp 3 parking is controlled by TA and has four aircraft parking spots, numbered 3-1 through 3-4. Spots 3-1 and 3-2 are marked to accommodate aircraft with a wingspan of 132 feet, 7 inches or less; Spots 3-3 and 3-4 are marked to accommodate aircraft with a wingspan of 117 feet, 5 inches or less. When requested, TA will inform tower of parking locations for all transient

aircraft and base assigned aircraft transporting Distinguished Visitors (DV). Spot 3-1 will normally only be utilized to enplane/deplane passengers or service the aircraft. The amount of time an aircraft is located on Spot 3-1 normally should not exceed 1 hour and 15 minutes. If mission requirements dictate additional time, users will coordinate with TA for parking at an alternate location and then will double-block back to Ramp 3 as necessary. See **Attachment 4**, Airfield Parking Ramps 1, 2, 3, 4.

1.10.1.4. Ramp 4 parking is controlled by TA and has six aircraft parking spots, numbered 4-1 through 4-6. Spots 4-1 and 4-2 are the designated primary parking for fighter aircraft without forward firing munitions. Spots 4-1 through 4-5 are designed southward-facing to accommodate C-130 aircraft; Spot 4-6 is designed to accommodate aircraft with a wingspan of 117 feet 5 inches or less (C-40). A variety of transient aircraft may park on the ramp. However, marshallers and wing walkers must be provided as necessary. See **Attachment 5**, Airfield Parking Ramps 4 and 5.

1.10.1.5. Ramp 5 parking is controlled by the 521st Air Mobility Operations Wing (521 AMOW) (Air Mobility Command [AMC]) through 721 AMOG. 721st Aircraft Maintenance Squadron Maintenance Operations Center (721 AMXS/MOC) is the ramp coordinating authority for its use. Ramp 5 is intended for the primary use of 18 AF (AMC) controlled missions. Transient aircraft operating other than 18 AF (AMC) controlled missions may be accommodated on a space-available basis with appropriate prior coordination through 721 AMXS/MOC. Use of Ramp 5 requires a specific Prior Permission Required (PPR) from 721 AMXS/MOC. This PPR is in addition to the PPR required/issued by AM for general use of Ramstein AB. Servicing of non-18 AF (AMC) aircraft shall be IAW local maintenance directives. Ramp 5's air terminal jetway parking spots are numbered T1 and T2. Spots T1 and T2 are nose-in spots, and aircraft must be pushed back in order to depart. Ramp 5 engine run Spots R1 and R2 require aircraft to be towed into place. Aircraft may taxi under power when departing R1 and R2.

1.10.1.5.1. Narrow-body (NB) aircraft are those with a wingspan of 170 feet or less and an overall length of 188 feet or less. Primary NB parking includes Spots 14-16, 20-21, and 23-28. Alternate NB parking spots are all parking locations.

1.10.1.5.2. Wide-body (WB) aircraft are those with a wingspan greater than 170 feet or an overall length greater than 188 feet. Primary WB spots are 4W, 5W, 6-13, 18-20 and 21W. Alternate WB overflow spots are 14W, 15W, and 17. The suffix "W" distinguishes spots from NB parking in the same location. Spots R1 and R2 can be used as nose-in spots for WB aircraft during Parking Maximum on Ground (Parking MOG) conditions. See **Attachment 5**, Airfield Parking Ramps 4 and 5.

1.10.1.5.3. See 721st Aircraft Maintenance Instruction (721 AMXSI 21-101), Maintenance Management, for individual parking spot restrictions for Ramp 5.

1.10.1.6. Ramp 7 (Southwest Area) is controlled by the 76th Airlift Squadron (76 AS) and has seven parking spots for C-21 aircraft. Additional aircraft parking is available utilizing hangars and dispersed aircraft shelters. See **Attachment 6**, Airfield Parking Ramps 7, 8.

1.10.1.7. Ramp 8/Hot Cargo Pad (HCP) is located on the south side of runway 08/26 at mid-field and is accessible via Taxiways Echo and Lima. 721 APS/TROC (Cape Forecasting) is the ramp coordinating authority. All aircraft requiring the use of Ramp 8 must coordinate with Cape Forecasting and 721 AMXS/MOC. Ramp 8 has five aircraft parking spots, although only three are usable at any one time. Ramp 8 can accommodate: two WB and one NB/three NB/one WB and

two NB. Spot 8-1 should be the last NB spot used to provide clearance for taxiing aircraft. Spot 8-2 and 8-3 cannot be utilized at the same time due to proximity of the spots. See **Attachment 6**, Airfield Parking Ramps 7, 8.

1.10.1.7.1. Aircraft with hazardous class 1.1 and 1.2 explosives must be parked on Ramp 8. Up to 3,000 pounds net explosive weight (NEW) of 1.3 is authorized on approved spots on Ramp 5. More than 3,000 pounds of NEW 1.3 must be parked on Ramp 8. Mission-essential quantity of hazardous Class 1.4 explosives is authorized on Ramp 5.

1.10.1.7.2. Use of Ramp 8 requires a specific PPR from 721 APS/TROC. This PPR is in addition to the PPR required/issued by AM or 721 AMXS/ MOC for general use of Ramstein AB. Ramp 8 PPRs are approved using Joint Chiefs of Staff (JCS) priority, munitions load, and projected availability at the time of the request. Higher priority munitions missions may subsequently bump down lower priority missions or non-munitions missions after PPRs are issued. In such cases, requestors will generally be notified for mitigation. In some cases, aircraft may be required to be downloaded and/or towed to accommodate higher priority munitions movements. Failure to coordinate a PPR with 721 APS/TROC may result in denied access during mission execution.

1.10.1.7.3. Aircraft block in/block out activities are controlled through 721 AMXS/MOC. Inbound aircraft, regardless of type, with a PPR for the HCP shall contact 86 AW CP NLT 30 minutes prior to arrival for final parking coordination. Unexpected aircraft diverts may be accommodated on a space available basis. However, divert aircraft may encounter parking delay until appropriate coordination can be accomplished and/or space becomes available. As much advance notice/coordination as possible will ensure the best possible accommodation.

1.10.1.8. Hardstand (HS)-131 and HS-132 (adjacent to Bldgs. 2525 and 2526) are designated to accommodate parking for aircraft with forward firing munitions. Contact Airfield Management (AM) for availability.

1.10.2. Deviation from marked parking plans should be forwarded to the Airfield Manager for coordination and approval at least 48 hours prior to a planned deviation. AM will inform tower of parking deviations.

1.11. Air Traffic Control Facilities. Ramstein has a VFR control tower and a GCA facility.

1.11.1. Ramstein Tower provides 24 hours-a-day VFR service within Ramstein's Class D airspace.

1.11.2. GCA provides ATC radar services within Ramstein Class E airspace from 0600L to 2200L, Monday through Saturday and from 1230L to 2200L on Sundays and German holidays. The GCA is closed on U.S. holidays and USAFE and 86 AW down days. IAW the Letter of Agreement (LOA) between Langen ACC and 86 OSS, the GCA will not control more than four aircraft simultaneously. Formations count as one "aircraft" when applying this rule. See **Attachment 7**, GCA Radar Pattern. Langen ACC controls all IFR departures except for aircraft that will remain within Ramstein GCA's airspace and departures during Tactical Air Navigation (TACAN) outages.

1.11.3. Primary Tower Visual/Radio Blind Spots. Ramps 1 and 7 are visual blind spots and the airfield has no radio blind spots.

1.12. Local Frequencies.

1.12.1. ATC must not authorize use of ATC frequencies by non-ATC users. Aircraft will use VHF (primary) when equipped to communicate with ATC in the local area.

1.12.2. All vehicles and ground personnel will use the Ramp Net to contact tower.

1.12.3. The Fire Department is authorized to use the frequency 279.775 during airfield emergencies.

Table 1.1. Local ATC Frequencies.

Facility	VHF	UHF
Ramstein GCA	140.9/124.280	284.675
Ramstein Tower	133.205	356.85
Ramstein Ground	121.775	300.65
Langen ACC	129.675	291.0

1.13. Radar, Airfield, and Weather Systems.

1.13.1. Ramstein TACAN (RMS). Channel 81; located at N 49˚ 26.08’ and E 07˚ 35.14’.

1.13.2. Instrument Landing System (ILS).

1.13.2.1. ILS Runway 08. CAT I.

1.13.2.2. ILS Runway 26. CAT I, II, and IIIa.

1.13.2.3. ILS Runway 09. CAT I.

1.13.2.4. ILS Runway 27. CAT I.

1.13.3. Digital Airport Surveillance Radar (DASR). The antenna that provides the Ramstein radar feed is located approximately 1 NM north-northwest of the runways.

1.13.4. TACAN checkpoints on the airfield are marked and located:

1.13.4.1. At the instrument hold lines for Taxiway Echo North (R-061/0.5 Distance- Measuring Equipment [DME]).

1.13.4.2. South of the instrument hold lines for Taxiway Echo South (R-99/0.6 DME).

1.13.5. RAWS preventative maintenance inspection (PMI) Schedule. German Daylight-Saving Time begins the last Sunday in March and ends the last Sunday in October.

Table 1.2. RAWS PMI Schedule.

System	Local	Non-Daylight Savings Time	Daylight Savings Time
DASR, ILS, TACAN	Monday – Sunday 2200L – 0530L	2100Z – 0430Z	2000Z – 0330Z

1.13.6. Current/forecasted weather requirements for downtime release: Ceiling 1,000 feet (304 meters) and visibility 3 statute miles (5,000 meters). Regardless of weather conditions, the ATC tower watch supervisor may release the DASR and Standard Terminal Automation Replacement System (STARS) when the GCA is closed.

1.14. Transient Alert Services. TA provides full service during airfield operating hours. Limited services are available for unscheduled arrivals and during airfield quiet hours. For additional information on TA services available at Ramstein AB, see Department of Defense (DoD) Flight Information Publication (FLIP) Enroute Supplement.

1.15. Automated Terminal Information Service. Tower will broadcast the ATIS IAW FAAO JO 7110.65 and local operating procedures, Monday through Saturday from 0500L to 2200L and Sundays and German holidays from 1200L to 2200L. During quiet hours, a standardized ATIS message will be broadcast. One hour prior to quiet hours ending, the regular ATIS will be broadcast. See **Attachment 13**, Consolidated Numbers and Frequencies for ATIS frequencies.

1.16. Aircraft Special Operations Areas/Ramps.

1.16.1. Arm/De-Arm, Hot Brake, Hung Flare, and Hydrazine Areas. Taxiways Alpha and Kilo are the primary arm/de-arm, hot brake, hung flare, and hydrazine areas. Taxiway Golf between Runways 09/27 and 08/26 is the secondary arm/de-arm, hot brake, hung flare, and hydrazine areas. See **Attachment 2**, Airfield Diagram and Controlled Movement Area (CMA) for depiction.

1.16.1.1. Aircraft requiring arm/de-arm or with hung/unsafe ordnance parked on Taxiway Kilo and Taxiway Golf center will be instructed to park facing due west (270-degree heading). Aircraft parking on Taxiway Alpha will be instructed to park facing southeast (120-degree heading).

1.16.1.2. Fighter aircraft requiring a hydrazine response will be instructed to park with the nose of the aircraft facing into the wind.

1.16.1.3. Fighters with forward firing munitions will position on Taxiway Kilo or Taxiway Alpha for arm/de-arm procedures. A maximum of four forward firing fighter aircraft permitted on the ground for Taxiway Kilo and six for Taxiway Alpha. These aircraft will be oriented with their nose facing south. Both locations require an event waiver through 86 AW/SEW and a contingency waiver through the Airfield Operations Flight (86 OSS/OSA).

1.16.1.4. The primary hydrazine servicing areas will be outside of Bldg. 2525 and the hot cargo pad.

1.16.2. Drag Chute Jettison Areas.

1.16.2.1. Aircraft equipped with drag chutes shall retain chutes until parked unless otherwise approved by tower.

1.16.2.2. Tower will inform AM where the drag chute is released, and AM will retrieve the chute. Tower or AM will suspend runway operations if necessary.

1.16.3. Hot Pit Refueling Areas/Alternate Fuel Cell Location.

1.16.3.1. IAW the Base Hot Refuel Certification Team Memorandum (2013), hot pit refueling is authorized on Ramp 5, Spot 14W, for 86 AW/XP special missions hot refueling operations only.

1.16.3.2. The C-130 alternate fuel cell is located on Ramp 1 Spot 14.

1.16.4. Large Aircraft Infrared Countermeasures (LAIRCM) Functional Tests. Aircraft self-defense system functional tests may be intentionally activated on Ramp 1 Spot 18 and Ramp 5 Spot 17 to confirm functionality and laser beam alignment. This must be coordinated with AM prior to testing beginning. All other locations shall be pre-coordinated at least 48 hours in advance and approved on a case-by-case basis by the Airfield Manager or Airfield Operations Flight Commander (AOF/CC). When notified:

1.16.4.1. AM shall:

1.16.4.1.1. Obtain start time, projected completion time, location of the test and confirm Ramp Net capable throughout the test.

1.16.4.1.2. Notify tower, Security Forces and Unit Airfield Driving Program managers of all pertinent info.

1.16.4.1.3. For testing conducted on Ramp 1, suspend aircraft taxi operations in front of Ramp 1, Spot 18 from Ramp 1 Hot Spot 2 to Ramp 1, Spot 17.

1.16.4.1.4. Notify tower, Security Forces and Unit Airfield Driving Program Managers when testing is complete.

1.16.4.1.5. Annotate start time, completion time and airfield restrictions due to LAIRCM testing and functional checks on AF Form 3616, Daily Record of Facility Operation.

1.16.4.2. Tower shall:

1.16.4.2.1. Ensure all coordination for LAIRCM testing is coordinated through AM.

1.16.4.2.2. Maintain direct communications with LAIRCM safety observers via the Ramp Net and request to be notified of all laser activations and termination activity.

1.16.4.2.3. Prohibit aircraft from taxiing on Ramp 1, in front of Spot 18 from Ramp 1 Hot Spot 2 to Ramp 1, Spot 17 while the laser is activated.

1.16.4.2.4. If notified by any other agency other than AM that LAIRCM testing is complete, inform AM.

1.16.4.2.5. Document start and stop times on AF Form 3616, Daily Record of Facility Operation.

1.16.4.3. Hazards and Safety. The LAIRCM pointer tracker and laser can radiate harmful levels of radiation up to 171 feet. Tower controllers shall not use binoculars to scan the surface area in the direction of LAIRCM testing. AM personnel shall remain outside of the safety zone and avoid using binoculars in the area of the test. To the maximum extent possible, all scheduled tests on Ramp 1 shall be conducted during quiet hours when there is minimal movement in the vicinity.

1.16.5. Recovering Aircraft Experiencing In-Flight Emergencies (IFE). Taxiways Lima and Echo (south of Runway 8/26) may be used for recovering aircraft experiencing in-flight emergencies to avoid congesting taxiways leading into Ramps 1-5.

1.17. Aircraft Towing Procedures.

1.17.1. The 86 MXG/MXOC (MOC) and 721 AMXS/MOC will coordinate all aircraft tow requests directly with the tower via landline. Exception: 76 AS C-21 maintainers do not have to call tower when putting C-21 aircraft into/out of hangars.

1.17.2. The MOC will not authorize towing and/or parking of aircraft on any taxiway without prior coordination and approval from AM. AM will notify tower of taxiway closures and send appropriate NOTAM.

1.17.3. Tow supervisors will ensure:

1.17.3.1. Ramstein Ground is contacted for approval when ready to begin tow operation.

1.17.3.2. All instructions are read back verbatim.

1.17.3.3. Ramstein Ground frequency is monitored at all times after initial contact and until tow operation terminates.

1.17.3.4. Ramstein Ground is advised when tow operation is complete.

1.18. Aircraft Taxiing Requirements/Routes.

1.18.1. General Procedures.

1.18.1.1. All aircraft will call Ground Control with current ATIS to request permission to taxi.

1.18.1.2. To reduce Foreign Object Damage (FOD), all aircraft can expect to make a 180- degree turn on the respective runways following the preferred direction: Runway 08/26, make a 180 degree turn toward the north; Runway 09/27, make a 180-degree turn towards the South. The 180-degree turn will only be executed on the concrete portions of the runway. Exception: C-21 and smaller aircraft are authorized to make 180-degree turns on all runways when the outside air temperature is less than 23 degrees Centigrade.

1.18.1.3. See max power engine run procedures in **Table 1.7**. Aircraft shall use caution when taxiing behind heavy jet aircraft.

1.18.2. Heavy Aircraft Jet Thrust Avoidance Procedures. Vehicles or pedestrians will follow procedures IAW AFI 13-213_RAMSTEINSUP, Airfield Driving, for jet thrust avoidance.

1.18.3. Hazardous Cargo Taxi Procedures. Unless directed otherwise, all hazardous cargo aircraft will back-taxi on the runway.

1.18.4. Aircraft Taxi During Emergencies. Vehicles must give way to emergency aircraft and responding emergency vehicles.

1.18.5. Taxiway/Taxilane Restrictions and Wingtip Clearance.

1.18.5.1. Taxiway Alpha. Unrestricted.

1.18.5.2. Taxiway Bravo. Unrestricted.

1.18.5.3. Taxiway Charlie. Restricted to aircraft with a wingspan of 142 feet or less (e.g., C-32, C-40, C-130, KC-135, etc.). Restriction is based upon obstacle clearance of 121 feet.

1.18.5.4. Taxiway Delta.

1.18.5.4.1. Taxiway Delta (East - Between Taxiway Charlie and Ramp 3): Restricted to aircraft with a wingspan of 142 feet or less (e.g., C-32, C-40, C-130, KC-135, etc.). Restriction is based upon obstacle clearance of 121 feet.

1.18.5.4.2. Taxiway Delta (West - Between Taxiway Echo and Ramp 2)): Restricted to aircraft with a wingspan of 170 feet or less (e.g., C-17, C-32, C-40, C-130, KC-135, etc.). Restriction is based upon obstacle clearance of 135 feet.

1.18.5.5. Taxiway Echo.

1.18.5.5.1. Taxiway Echo (South): Unrestricted.

1.18.5.5.2. Taxiway Echo (Middle): Unrestricted. Note: Holding aircraft with a length of 154 feet or more on Taxiway Echo Middle will occupy the vacated runway. Examples of aircraft that are larger than 154 feet: C-5, C-17, KC-10, B-747, B-767, B- 777.

1.18.5.5.3. Taxiway Echo (North): Unrestricted.

1.18.5.6. Taxiway Foxtrot. Restricted to aircraft with a wingspan of 222 feet 9 inches or less (e.g., C-5, E-4, VC-25, B-747, B-767, B-777, etc.). Restriction is based upon obstacle clearance of 161 feet 5 inches.

1.18.5.7. Taxiway Golf.

1.18.5.7.1. Taxiway Golf (South): Restricted to aircraft with a wingspan of 78 feet or less (e.g., A-10, C-12, C-20, C-21, etc.). Restriction is based upon obstacle clearance of 68 feet 10 inches. Base assigned C-130 and C-37 are only authorized to taxi through this area if approval is received from the Airfield Manager.

1.18.5.7.1.1. Ramp 7, through Taxilane (East): Restricted to aircraft with a wingspan of 48 feet or less (e.g., C-21, etc.) when Spot 7-3A is occupied; 77 feet 10 inches or less (e.g., C-12, C-20, C-21, etc.) when Spot 7-3 and/or 7-6 is occupied.

1.18.5.7.1.2. Ramp 7, through Taxilane (West): Restricted to aircraft with a wingspan of 64 feet or less (e.g., C-12, C-21, etc.) when Spot 7-3 and/or Spot 7-4 is occupied.

1.18.5.7.1.3. Southern Taxitracks: Restricted to aircraft with a wingspan of 39 feet 6 inches or less (e.g., C-21); Day/VFR operations only unless operating under NVD conditions.

1.18.5.7.2. Taxiway Golf (Middle): Unrestricted.

1.18.5.7.3. Taxiway Golf (North): Restricted to aircraft with a wingspan of 224 feet 3 inches or less (e.g., C-5, E-4, VC-25, B-747, B-767, B-777, etc.). Restriction based upon obstacle clearance of 162 feet 6 inches.aero

1.18.5.8. Taxiway Kilo. Unrestricted.

1.18.5.9. Taxiway Lima. Unrestricted.

1.19. Airfield Maintenance.

1.19.1. Sweeper Operations.

1.19.1.1. During the period of 14 Apr – 14 Nov, unless weather conditions warrant otherwise, 786 CES/CEOHP (Heavy Equipment) shall assign one sweeper with a qualified operator to the airfield Monday – Friday, 0700L – 1600L, except on U.S. holidays and USAFE 86 AW down days, etc. During non-duty hours and holidays/morale days, 786 CES/CEOHP shall have an individual on stand-by, available to respond to the airfield environment with a sweeper within 1 hour of notification. Contact 786 Civil Engineer Squadron Customer Service desk during non-duty hours for standby sweeper.

1.19.1.2. During the snow season, 15 Nov – 15 Apr, unless weather conditions warrant otherwise, 786 CES/CEOHP will be manned 24/7 to provide an immediate sweeper response capability.

1.19.1.3. The following airfield sweeper schedule is established and carried out starting at 0700L each duty day. AM will contact sweeper via radio or landline for additional sweeping requests throughout the day.

Table 1.3. Airfield Sweeper Schedule.

Monday	Runway 08/26, Taxiways Golf, Echo, Foxtrot, Lima, & Kilo, Ramps 5, 7, & 8
Tuesday	Runway 09/27, Taxiways Alpha, Bravo, Charlie, Delta, Echo, & Foxtrot, Ramps 1, 2, 3, & 4
Wednesday	Airfield Access Points & the northern access road
Thursday	Runway 08/26, Taxiways Golf, Echo, Foxtrot, Lima, & Kilo, Ramps 5, 7 & 8
Friday	Runway 09/27 , Taxiways Alpha, Bravo, Charlie, Delta, Echo, & Foxtrot, Ramps 1, 2, 3, & 4
Mon - Fri 0800L, 1300L & 1545L	Mitchell Avenue between 86 AW HQ and the northern access road

1.19.1.4. The airfield sweeper operator will notify AM via landline (see **Attachment 13**, Consolidated Numbers and Frequencies for AM number) or radio (Ramp Net) when daily sweeping operations have begun and when they are completed. AM will log the airfield sweeper on the airfield, all sweeper requests, and completion notifications on AF IMT 3616, Daily Records of Facility Operations IAW Air Force Manual (AFMAN) 13-204v3 USAFE Supplement.

1.19.2. Grass mowing is scheduled IAW the Wing BASH Plan. Requirements outside of this schedule are coordinated with AM.

1.19.3. Host Nation Farming. Ramstein AB does not have any host nation farming agreements.

1.20. Runway Surface Condition (RSC) and/or Runway Condition Reading (RCR) Values.

1.20.1. AM will accomplish RCR/RSC checks IAW AFMAN 13-204v3, Airfield Operation Procedures and Programs and TO 33-1-23, Equipment and Procedures for Obtaining Runway Condition Readings. AM will report RCR/RSC values to the following agencies: Tower, GCA, 86 AW/CP, Weather (86 OSS/OSW), 786 CES Snow Control Center (during winter/snow season), and 721 MSS/AMCC. AM is the only agency authorized to determine the RSC. Tower and AM are the only agencies authorized to report the RSC.

1.20.2. Upon request from ICAO aircraft, tower and GCA will convert RCR readings according to Flight Information Handbook, using **Table 1.4**.

1.20.3. Reportable RSCs are: Dry, Wet, Loose Snow, Packed Snow, and Ice.

Table 1.4. Braking Coefficient and ICAO Conversion.

RCR Value	ICAO Index
>17	5
15-17	4
10-14	3
5-9	2
0-4	Not used

Table 1.5. Braking Action Conversion.

Pilot Reported Braking Action	RCR Value
Good	14-22
Good to Medium	13
Medium	10-12
Medium to Poor	8-9
Poor	7
Nil	0-6
Note: The above tables are derived from the Flight Information Handbook.	

1.21. Procedures/Requirements for Conducting Runway Inspections/Checks.

1.21.1. AM will conduct an airfield inspection at least once daily.

1.21.2. AM will conduct and document additional checks to examine the primary takeoff, landing and taxi surfaces, when necessary, in support of events that could cause unsafe airfield conditions.

1.21.3. 786 CES/CEOF (Airfield Lighting) will conduct an airfield lighting check at least once daily Monday through Friday, excluding holidays (German and US). This inspection will include all lights associated on the airfield and the SFLs located outside the base perimeter fence. Airfield lighting will report the status of airfield lighting systems to AM upon completion. On holidays (German and US) and weekends, AM will perform the lighting check including the sequencing flashing lights located outside the base perimeter fence.

1.21.4. Civil engineering construction (86 CES/CEPM) will inspect areas closed for repair/construction prior to opening in conjunction with AM and 86 AW/SE (Wing Safety).

1.21.5. The 786 CES/CEOF will check and report the status of the arresting barriers to AM daily. Status will be reported to AM as “operational or nonoperational.”

1.22. Procedures for Opening and Closing the Runway.

1.22.1. AM is the approval authority for opening/closing runways.

1.22.2. When a situation occurs where a runway needs to be closed, AM will call the runway closed and send a NOTAM. AM will contact the following agencies and give an estimated time frame when the runway is expected to open: Tower, 86 AW/CP, 721 MSS, 721 AMXS, 786 CEO/CEOFP, 86 MXG/MXOC (MOC), 86 MXS/TA (TA), Fire Department, and 86 AW Safety.

1.22.3. AM will re-open the runway after it is checked to ensure that it is free of all hazards (aircraft, vehicle, debris, etc.). AM will notify the agencies in **Paragraph 1.22.2** when the runway is open and will cancel applicable NOTAM.

1.22.4. Scheduled airfield/runway closures will be approved in accordance with the AFMAN13-204 USAFE Supplement AM will publish a NOTAM and coordinate the information with the following agencies and give an estimated time frame when the runway is expected to resume ops: Air Traffic Control, 86 AW/CP, 721 MSS, 721 AMXS, 86 MXG/MXOC (MOC), 86 MXS/TA (TA), Fire Department, and 86 AW Safety.

1.23. Procedures for Suspending and Resuming Runway Operations.

1.23.1. IAW AFMAN 13-204v3 USAFE Supplement, runway operations may be suspended at the discretion of AM or the tower watch supervisor/senior controller.

1.23.2. When a situation occurs where runway operations need to be suspended, AM will suspend runway operations and send a NOTAM (if appropriate). Situations that may drive a suspension include, but are not limited to, construction, snow removal, hazardous weather conditions, or damage to landing surfaces.

1.23.3. Operations are automatically suspended on the affected runway after an emergency aircraft has landed, except for physiological or medical emergencies, and remain suspended until AM has inspected the runway and informed tower that runway operations may resume. Note: When runway operations are suspended, all vehicles must continue to obtain clearance from the tower prior to entering or crossing the runway.

1.23.4. When the suspension is directed by the tower, the watch supervisor/senior controller will inform AM immediately when suspending runway operations. Only AM may resume runway operations following suspension or closure.

1.23.5. Upon suspension, AM will contact the following agencies and give an estimated time frame when the runway is expected to resume operations: Tower, 86 AW/CP, 721 MSS , 721 AMXS, 786 CEO/CEOFP, 86 MXG/MXOC (MOC), 86 MXS/TA (TA), Fire Department, and 86 AW Safety. AM will contact the following agencies and give an estimated time frame when runway operations are expected to resume: Tower, 86 AW CP, 721st Mission Support Squadron (721 MSS), 86 MXG/MXOC (MOC), 86 MXS/TA (TA), Fire Department and 86 AW Safety.

1.23.6. When operations are resumed, AM will notify Tower, 86 AW CP, 721 MSS, 86 MXG/MXOC (MOC), TA, Fire Department, and 86 AW Safety.

1.24. Noise Abatement Procedures. To minimize noise pollution for Ramstein AB and surrounding communities, the following procedures and restrictions apply:

1.24.1. Fighter aircraft will not depart VFR with ceiling less than 3,700 feet AGL and visibility less than 5,000 meters.

1.24.2. Reverse thrust. Aircraft landing are requested to limit the use of reverse thrust to the maximum extent possible consistent with safety of flight. Tactical landing training is exempt.

1.24.3. Approach procedures. IAW with FLIP Area Planning (AP)/2, pilots should arrange their flight to leave the Initial Approach Fix (IAF) at a speed which permits operation of the aircraft in clean configuration. This speed should be maintained until reaching a distance of 12 NM from touchdown. For this portion of the approach, an indicated airspeed of 210 +/- 10 knots is recommended, unless a higher airspeed is required for performance reasons. Landing configuration should be established shortly prior to or over the final approach fix.

1.24.4. Visual approaches should avoid overflight of towns and villages. When approaching from the east, pilots should execute the Autobahn 6 Visual Approach for Runway 26 or 27 (See FLIP).

1.24.5. Aircraft will avoid overflying villages to the maximum extent possible, except as required in an emergency, or when executing a departure procedure.

1.24.6. Avoidance of USAFE HQ and 86 AW/CC buildings. To the maximum extent possible, aircraft will avoid overflight of the areas on Ramstein AB encompassing the USAFE headquarters

and the 86 AW/CC building below 3,600 feet. Arriving VFR aircraft will be instructed to land via runway or taxiways in such a manner that avoids this area.

1.24.7. When crossing the airport north/south-bound, transient VFR aircraft will be instructed to cross the east or west ends of the runway and avoid midfield overflight. Transient VFR aircraft crossing the Class D surface area east/west bound will remain south or four miles north of the runway until they have passed the departure end of the runway for their direction of flight. Pipeline inspection helicopters, federal agency, and emergency response operations aircraft are exempt from this restriction.

1.24.8. Aircraft operations during quiet hours. The following general guidance will be applied to reduce noise pollution during quiet hours without negatively impacting the mission:

1.24.9. Aircrews and maintainers will operate their aircraft/engines to produce the least amount of noise consistent with safety of flight/mission accomplishment.

1.24.9.1. During quiet hours, pilots with quiet hour approval are only authorized one instrument approach to a full stop landing and/or one departure.

1.24.9.2. Aircraft are not authorized IFR/VFR patterns on weekends and German holidays except in emergency situations or upon approval by the 86 OG/CC.

1.24.9.3. When quiet hour operations are authorized, tower will advise pilots to avoid overflight of populated areas.

1.25. Quiet Hour Procedures/Waivers.

1.25.1. Ramstein AB maintains strict quiet hours for flight operations every day. All users are required to comply with these arrival/departure quiet hours restrictions or request a waiver. Ramstein AB quiet hours for aircraft arrivals and departures are:

1.25.1.1. Quiet Hours begin every evening at 2200L. Quiet Hours end at 0600L Monday through Saturday and 1300L on Sundays and German holidays. Engine starts are not approved without a Quiet Hour Waiver (QHW) before 0600L Monday through Saturday or 1230L on Sundays and German holidays.

1.25.1.2. Tower personnel may taxi aircraft to parking that land at or before 2200L.

1.25.2. Training flights are not authorized on weekends or German holidays. This does not prevent 86 AW-assigned units from arriving to or departing from Ramstein AB on training flights if they accomplish no practice approaches (i.e., strictly positioning and depositioning for an off-station trainer).

1.25.3. The 86 OG/CC (or acting) is the approval authority for arrival and departure QHW requests. AM is the delegated authority and may approve up to six (6) arrival/departure QHW requests per day for any missions (86 AW-assigned or transient aircraft) that fall under one or more of the following categories:

1.25.3.1. CJCS mission priority 1A1, 1A2, 1A3, or 1B1.

1.25.3.2. Patriot Express mission operating in delay.

1.25.3.3. Aeromedical evacuation.

1.25.3.4. Repatriation of human remains.

1.25.3.5. Diverts to Ramstein AB due to weather or in-flight emergency.

1.25.4. The following quiet hours apply to aircraft engine runs.

1.25.4.1. Monday through Saturday: 0001-0700L and 1900-2359L. Engine runs during these windows require a QHW.

1.25.4.2. Sundays and German holidays: 0001-2359L. Engine runs on Sundays and German holidays require a QHW.

1.25.5. The 86 MXG/CC (or acting) is the approval authority for engine run QHW requests for 86 AW-assigned aircraft and transient aircraft not controlled by TACC operating at Ramstein AB. 86 AMXS or 86 MXS Production Superintendent is delegated authority to approve engine run QHW requests for 86 AW-assigned aircraft with all engines at ground idle.

1.25.6. The 86 OG/CC (or acting) is the approval authority for engine run QHW requests for transient TACC-controlled aircraft operating at Ramstein AB. Engine runs will only occur on Ramps 1, 2, 3, 4, 5, and 7. AM is delegated authority to approve up to four (4) engine run QHW requests for any TACC-controlled transient aircraft at ground idle power that are executing a mission that falls under one or more of the following categories:

1.25.6.1. CJCS mission priority 1A1, 1A2, 1A3, and 1B1.

1.25.6.2. Patriot Express mission operating in delay.

1.25.6.3. Aeromedical evacuation.

1.25.6.4. Repatriation of human remains.

1.25.7. Maintenance engine runs up to any power for aircraft supporting the high priority missions listed below will be IAW the German regulation titled Approval Required under Air Traffic Law for the Extension of Ramstein Military Airbase (will be referred to as "ATAP" [Air Traffic Act Permit] for the remainder of the document), meaning no more than two engines at a time.

1.25.7.1. 1A1, 1A2, 1A3 missions.

1.25.7.2. Urgent and priority aeromedical evacuations.

1.25.7.3. Repatriation of human remains.

1.25.8. C-17, C-5M, and civilian aircraft that are "Stage 4" compliant may accomplish maintenance engine runs at idle power from 2200L – 0700L or on German holidays and Sundays for priority 1B1 missions departing ETAR for Area of Responsibility (AOR)-bound contingency deployments.

1.25.9. Tower and AM will monitor airfield operations during both arrival/departure and engine run quiet hours and notify 86 AW/CP of any deviations.

1.25.10. AM is the central coordinator for all QHW requests. The QHW SharePoint (**https://usaf.dps.mil/sites/86_OG/QHW/**) is the primary means for submitting QHW requests. Requestors must submit for two distinct QHWs if an aircraft's arrival and departure will both violate quiet hours.

1.25.11. If planners/mission managers desiring to submit a QHW request do not have access to the QHW SharePoint site, they may e-mail the 86 OG/CCE shared mailbox (**86og.cce@us.af.mil**) to request access. The alternate means for submitting QHW requests is by e-mailing a completed

"Ramstein AB QHW Request Form" (see Attachment 15) to AM (**86oss.osaaairfieldmanagementoperations@us.af.mil**), who will in-turn transfer the respective info to QHW SharePoint website.

1.25.12. When submitting a QHW request, planners and mission managers should provide all available mission information and justification to inform the approval authority's decision making. Consideration will be given to mission requirements and planning constraints, such as diplomatic clearances, aircraft availability, passenger timelines, and parking availability.

1.25.13. AM (arrivals, departures, and engine runs for 618 AOC-controlled) and 86 MXG/MOC (all others) will coordinate all QHW requests with the appropriate approval authority and update the QHW SharePoint website with approved/denied status. If the QHW approval authority is the 86 OG/CC or the 86 MXG/CC and the QHW request is urgent (i.e., cannot wait until normal business hours for staffing), AM will contact 86 AW/CP to be connected with the applicable on-station QHW approval authority. Users requesting a QHW can confirm approved/denied status by checking the QHW Site. AM will directly inform those who submitted the manual form of the approved/denied decision.

1.25.14. An approved QHW is valid through that day's entire quite hour period unless the approval authority imposes specific restrictions.

1.26. Engine Test/Run-Up Procedures.

1.26.1. IAW with the German ATAP, during engine test runs, only two engines per aircraft are allowed to be operated simultaneously under load conditions, unless necessary for fulfillment of mission-related tasks. For guidance on engine runs that must take place during quiet hours, please see section 1.24. Quiet Hour Procedures/Waivers.

1.26.2. The 86 MXG/MXOC (MOC) and 721 AMXS/MOC will coordinate all engine runs with tower. Engine run supervisors will ensure ground control is contacted for approval prior to engine start. All non-86 AW or non-AMC assigned aircraft will coordinate engine runs through the 86 MXG/MXOC (MOC) 86 MXG/MXOC will notify 721 AMXS of the engine run operations of non-AMC assigned aircraft parked on Ramp 8 and Ramp 5.

1.26.3. Once approval is given, maintenance engine run supervisors will continue to monitor ground control. Aircraft radio transmissions to ground control will be kept to a minimum.

1.26.4. Rules for aircraft engine runs on Ramps 5 and 8 are listed in **Table 1.6**. Rules for engine runs by base assigned aircraft are listed in **Table 1.7**.

Table 1.6. Authorized Engine Run Spots on Ramps 5 and 8 (Except C-130s).

Rule	Conditions
1	Idle runs may be accomplished on any parking location.
2	Engine runs requiring two engines above idle will use the designated run Spots 5 (facing east), R1 and R2 (facing north) unless spots are unavailable. In this case, refer to Rule 3 (See Note 1).
3	Engine runs requiring two engines above idle may use Spots 5-6 through 5-11 and Spots 5-18 through 5-24 (facing west) provided two spots behind the aircraft are vacant. See Note 2.

4	C-17 reverse engine run-ups may be accomplished on all spots except T1 and T2. See Note 3.
Note 1: All aircraft with a wingspan greater than 169 feet 10 inches (C-17) to be parked on R1 or R2 will be pushed back onto parking spots.	
Note 2: Any maintenance runs must be coordinated through 86 MXG MOC, 721 AMXS/MOC, and the ATOC. Spotters will be positioned to ensure nothing passes behind the aircraft. All runs described in Table 1.3 will have 1000-feet clearance and guards facing the direction of the run.	
Note 3: All engine runs procedures to include power levels and direction, are the responsibility of the 721 AMXS and/or 86 MXG. Aircrew members or engine run supervisors assume full responsibility for the safety of all surrounding personnel and equipment. Tower serves only to approve engine runs for anti-hijack purposes. 86 OSS/OSA takes no responsibility for incorrect engine run procedures.	
Note 4: Reverse engine runs are prohibited while nacelle is positioned over unsecured fuel hydrant pit covers or manhole covers.	

Table 1.7. Authorized Engine Run Spots for Base Assigned Aircraft.

Ramp 1	
Rule	Conditions
1	Any spot for idle engine runs.
2	C-130 Aircraft (All): Max power engine runs are authorized on parking spots provided there is a blast deflector or another spot directly behind vacant of personnel, equipment and aircraft. The spot two rows behind, if applicable, will be vacant of non-powered Aerospace Ground Equipment AGE; all personnel will be restricted from upper surfaces and aircraft, if present, will be in the high winds configuration. The running aircraft's nose will face west, in alignment with nose wheel block/centerline. (See Notes 1 and 2)
Note 1: Safety spotters will be positioned to ensure nothing passes within 1,000 feet behind the aircraft.	
Note 2: Any maintenance runs on these spots must be coordinated through 86 MXG MOC, 721 AMXS/MOC, and the ATOC.	
Note 3: C-130 engine compressor washes can only be accomplished on Ramp 1 using a wash containment system. If a wash containment system is not available, engine compressor washes are prohibited.	
SOUTHWEST AREA (Ramp 7)	
1	C-21 engine runs authorized at HS-91 only.
Note 4: All aircraft are authorized engine runs to include full power on the hammerheads for minor maintenance problems only.	
Note 5: All engine run procedures to include power and direction, are the responsibility of the 86 MXG. Aircrew members or engine run supervisors assume full responsibility for the safety of all surround personnel and equipment. Tower serves only to approve engine runs for anti-hijack purposes. 86 OSS/OSA takes no responsibility for incorrect engine run procedures.	

1.27. Procedures for Protecting Precision Approach Critical Areas - ILS CAT I/II/IIIa.

1.27.1. During ILS CAT I/II/IIIa operations, separation between one approaching aircraft and another approaching or departing aircraft shall be provided in such a way as to ensure that approaching and departing aircraft, taxiing aircraft, or vehicles do not cause any interference of the ILS signal when the approaching aircraft is within a distance of 2 NM of the runway threshold. When the DASR and/or Tower Display Workstation is inoperative during Instrument Meteorological Conditions (IMC) or Low Visibility Conditions (not to be confused with "VISCONS"; see definition in **Attachment 1**), the approaching aircraft distance is increased to a distance of 4 NM of the runway threshold IAW Zentralrichtlinie Militärisches Flugverkehrsmanagement A2-272/2-2000-14. See **Attachment 9**, ILS CAT I/II/IIIa Critical Areas. Note: During ILS CAT I/II, a preceding aircraft approaching the same runway, or another runway may pass through the critical area while landing, departing, or exiting the runway; do not allow aircraft to stop within the critical areas IAW AFMAN 13-204v3.

1.27.1.1. All aircraft, vehicles and personnel will stop at the appropriate ILS Category instrument hold lines and contact tower for permission to enter the critical areas when CAT I/II/III procedures are in effect.

1.27.1.2. "Follow Me" service is available upon pilot request if visibility is too low to safely taxi in the judgement of the aircraft commander. In the event the lowest reported RVR value at any sensor falls below 200 m, "Follow-Me" service is required. Exception to this policy requires 86 OG/CC approval.

1.27.1.3. When operating in CAT II/IIIa, certain outages to airfield lighting systems may result in a downgrade to the ILS Category IAW Zentralrichtlinie Militärisches Flugverkehrsmanagement A2-272/2-2000-14. In such instances, AM will publish a NOTAM informing aircrews and will ring out the Secondary Crash Net (SCN) informing base agencies of the downgrade to available service.

1.27.2. ILS CAT I Operations.

1.27.2.1. ILS CAT I operations are in effect when the reported ceiling is less than 800 feet and/or the visibility is less than 2 miles.

1.27.2.2. All aircraft, vehicles and personnel shall be out of the CAT I ILS critical area prior to an approaching aircraft, executing an ILS approach, reaching 2 NM from the landing threshold (4 NM when the DASR is out of service).

1.27.3. ILS CAT II Operations.

1.27.3.1. ILS CAT II operations are in effect when the ceiling is less than 200 feet, but not lower than 100 feet, and/or the visibility is less than 800 m and/or the lowest reported RVR value at any sensor is less than 550m but not lower than 300 m.

1.27.3.2. During ILS CAT II/IIIa operations IAW Zentralrichtlinie Militärisches Flugverkehrsmanagement A2-272/2-2000-14:

1.27.3.2.1. The Runway 26 SFLs and PAPIs shall be switched off.

1.27.3.2.2. The runway centerline lights, the red side row barrettes and the taxiway centerline lights, and the stop bars shall be switched on.

1.27.3.3. Tower shall include on the ATIS IAW Zentralrichtlinie Militärisches Flugverkehrsmanagement A2-272/2-2000-14 the following statement, “LOW VISIBILITY PROCEDURES CAT II IN OPERATION.”

1.27.3.4. Tower shall broadcast a message on all available frequencies including the Ramp Net stating, “CAT II instrument hold procedures in effect."

1.27.4. ILS CAT IIIa Operations.

1.27.4.1. ILS CAT IIIa operations are in effect when the ceiling is less than 100 feet and/or the reported RVR value at any sensor is less than 300 m but not lower than 175 m.

1.27.4.2. The procedures stated in **Paragraph 1.27.3** for ILS CAT II operations remain in effect. In addition, tower shall include on the ATIS IAW Zentralrichtlinie Militärisches Flugverkehrsmanagement A2-272/2-2000-14 the following statement, “LOW VISIBILITY PROCEDURES CAT IIIa IN OPERATION.”

1.27.4.3. Ramstein tower shall broadcast a message on all available frequencies including the Ramp Net stating, “CAT IIIa instrument hold procedures in effect."

1.27.5. Measures to be taken for CAT II/IIIa Operations IAW Zentralrichtlinie Militärisches Flugverkehrsmanagement A2-272/2-2000-14. The following measures will be initiated when weather conditions are at or below a 300 ft ceiling and the RVR is less than or equal to 1000 m.

1.27.5.1. AM will make an initial CMA check (CAT II/IIIa critical and sensitive area) to ensure all vehicles, personnel, and/or aircraft in the CMA are accounted for and in contact with the tower prior to commencing ILS CAT II/IIIa operations. AM will assist tower in clearing/vacating all CAT II/IIIa critical and sensitive areas of all non- mission essential personnel, equipment and vehicles. AM shall report the status of CMA check to tower upon completion.

1.27.5.2. All aircraft, vehicles, and personnel are required to hold short at the CAT II/IIIa instrument hold lines upon receipt of notification from the tower.

1.27.5.3. The secondary power supply for optical (i.e. airfield lighting system) and non-optical landing aids (i.e. ILS) shall be verified as operational.

1.27.6. ILS CAT II/IIIa Restrictions.

1.27.6.1. Arrival vs. Arrival: The GCA will separate successive arriving aircraft by 10 NM in order to allow the lead aircraft to land, turn off the runway, and clear the ILS critical area.

1.27.6.2. Arrival vs. Departure: Departing aircraft will be separated from arriving aircraft on an ILS approach by ensuring the departing aircraft is departure roll prior to the ILS aircraft reaching 10 NM from the landing threshold.

1.28. Restricted/Classified Areas on the Airfield.

1.28.1. Restricted areas are established pursuant to DoD Instruction 5200.08, Security of DoD Installation and Resources and the DoD Physical Security Review Board (PSRB), and Section 21, Internal Security Act of 1950 Title 50 U.S.C. 797.

1.28.1.1. Permanent restricted areas on the airfield include Ramps 1, 2, 4, 5, and 8.

1.28.1.2. Other areas on the airfield may become restricted based on the protection level (PL) of the aircraft parked there. This will happen on a real-time basis.

1.28.2. All personnel must obtain specific written permission to enter restricted and controlled areas.

1.28.2.1. Written permission consists of an Automated Entry Control Card, commonly referred to as a line badge.

1.28.2.2. Personnel requesting an Automated Entry Control Card should contact their Unit Security Managers to initiate the process. After contacting one's Unit Security Manager and completing the appropriate documentation (provided by Security Manager), line badges may be obtained through the Security Forces Pass and Identification (ID) office. Transient personnel may use line badges issued at their home station IAW 86 AW local directives. Contact 86 Security Forces Pass and Identification for specific information on required documentation.

1.28.3. Restricted areas on and around the airfield are defined in 86 AW local directives.

1.28.3.1. Restricted areas are denoted by painted red lines.

1.28.3.2. Entry into a restricted area is permitted only through entry control points marked by painted white bars.

1.28.4. Questions regarding restricted/classified areas should be forwarded to 86 SFS/S5P and S5A.

1.29. Auxiliary Power for RAWS Facilities.

1.29.1. Battery power. All navigational aids (NAVAID) have backup batteries that provide up to one hour of power in the case of a commercial power failure. For those facilities equipped with backup generators, the battery systems are designed to provide power in the interim period between commercial power and generator power.

1.29.2. Generator power. Tower, GCA, TACAN, all ILSs, and DASR have diesel generators that can provide backup power in the event of a commercial power failure. Power Production (786 CES) is responsible for manual generator re-start in the case that the tower/AM generator fails to auto-start. Tower, GCA, and AM personnel will be trained annually to perform manual restart of their respective facilities' generators.

1.29.3. Uninterruptible power supply (UPS). GCA, tower, TACAN, and DASR are each equipped with an UPS system to provide power in the interim between commercial power interruption and the start of generator power.

1.30. Quiet Hour/Ramp Freeze Requests.

1.30.1. Quiet hours and ramp freezes may be requested for ceremonies and events taking place on Ramstein AB when noise reduction from aircraft operations, flight line ground support equipment, and flight line vehicles is required.

1.30.2. Airfield quiet hours restrict aircraft movement and noise for short periods of time in support of high-priority DV functions/movements. These quiet hour periods have the same restrictions/stipulations as Ramstein AB daily quiet hours as discussed in **Paragraph 1.23**.

1.30.3. A ramp freeze restricts aircraft and vehicle movement and noise on specific ramps and/or taxiways. Ramp freezes are usually in support of base functions (e.g., wing runs, change of command, etc.) and DV functions/movements.

1.30.4. To submit a request for quiet hour/ramp freeze, the requesting project officer will do the following:

1.30.4.1. Contact 86 AW/CCE and 86 AW/CCP (Protocol) for event checklist.

1.30.4.2. Schedule the event around any non-DV and non-AMC related arrivals and departures to the maximum extent possible. Contact 86 OSS/OSO for local aircraft movements and AM for transient non-AMC aircraft movements.

1.30.5. Coordination for 86 OG/CC approval of the quiet period or ramp freeze will be via a Staff Summary Sheet (AF Form 1768) or Electronic Staff Summary Sheet.

1.30.5.1. Include the following agencies in the routing block for coordination: 37 AS, 76 AS, 86 AMXS, 86 MXS, 86 AW/PA, 86 AW/CCP, 86 OSS/OSO, 86 OSS/OSA, 86 OG/CCE, and 86 OG/CC (approval).

1.30.5.2. Include the location on the airfield, date, and times, and requested operational restrictions (APU, engine run, taxiing, etc.)

1.30.5.3. Upon receipt of approval from the 86 OG/CC, AM will issue a NOTAM no earlier than 72 hours prior to the event.

Chapter 2

FLYING AREAS

2.1. Local Flying Area/Designation of Airspace.

2.1.1. Ramstein Class D airspace. Airspace measuring 14 nautical miles (NM) by 7 NM from the surface up to and including 3600 feet MSL: 4 NM north and 3 NM south of Runway 09/27 centerline, 7 NM east and 7 NM west centered on Runway 09/27. See **Attachment 8**, Class D Airspace/VFR Overhead/Inside/Outside Closed Pattern for depiction.

2.1.2. Ramstein GCA airspace. Class E airspace extending from the surface up to and including 5000 feet MSL, approximately 20 NM by 40 NM total, of which roughly 20 NM by 25 NM is controlled by Ramstein at any time, depending on active runway. See **Attachment 7**, GCA Radar Pattern for depiction.

2.2. VFR Local Training Areas.

2.2.1. An increase of glider operations can be expected during the summer months. Aircraft may be restricted from conducting full procedure approaches from MAPIG and XIDOD or expect to hold at MAPIG or XIDOD during periods of reported glider activity in vicinity of those fixes unless flying in VMC conditions above glider activity.

2.2.2. Aircraft receiving vectors to Runway 08 or 09 should request and expect short vectors to maintain maximum separation from MAPIG.

Chapter 3

VFR PROCEDURES

3.1. VFR Minimas.

3.1.1. IAW 14 CFR Part 91, VFR is in effect when the observed ceiling is greater than or equal to 1,000 feet Above Ground Level (AGL) and visibility is greater than or equal to 5,000 meters.

3.2. VFR Traffic Patterns. See **Attachment 8**, Class D Airspace/VFR Overhead/Inside/Outside Closed Pattern.

3.2.1. Class D airspace entry procedures, standard tower frequency/VFR reporting points.

3.2.1.1. Pilots must establish two-way radio communication with the tower and obtain approval for the requested pattern to be flown prior to entering Class D airspace.

3.2.1.2. The primary tower radio frequency for all aircraft conducting operations in Class D airspace is VHF 133.205.

3.2.1.3. The VFR reporting points are ECHO (R-062/5.2 DME), WISKY (R-291/6.6 DME) and PIVOT (R-352/6.6 DME). All radials and DME are from Ramstein's TACAN. Pilots must be familiar with the VFR entry/reporting points listed in **Attachment 8**, Class D Airspace/VFR Overhead/Inside/Outside Closed Pattern prior to working in Class D airspace. Unless otherwise coordinated, enter Class D airspace and proceed to the approved reporting point at or below 3,000 feet MSL (2,200 feet AGL).

3.2.1.4. Expect extensive glider activities in the vicinity of Ramstein AB during VFR conditions. Local gliders are not in communication with Ramstein GCA and are not transponder-equipped to provide traffic alert and collision avoidance system guidance. Aircrews should be extremely vigilant when operating in the vicinity of Ramstein AB.

3.2.1.5. All locally-assigned aircraft should exercise caution when traversing points WISKY, PIVOT, and ECHO in the local pattern. The Potzberg Hotel (Castle) performs aviary shows at their location; thus, the risk of striking a large predatory bird is increased over the area. Crews should plan to avoid this area (49˚31.20' N and 7˚28.80' E) by at least 0.5 NM.

3.2.2. Pattern use/deviations.

3.2.2.1. VFR patterns are authorized for use by all locally-assigned 86 AW aircraft, TDY aircraft operating under 86 AW ADCON, and NATO aircraft when training in conjunction (interflying) with the 37th Airlift Squadron. Transient aircraft may fly the VFR pattern when an abnormal or emergency situation exists or when authorized by the 86 OG/CC.

3.2.2.2. USAFE fighter aircraft are authorized to recover using the overhead pattern. However, aircraft desiring multiple approaches must execute their option back to the GCA radar pattern.

3.2.2.3. VFR patterns are closed during quiet hours, weekends, and German holidays.

3.2.3. Closed Traffic Patterns.

3.2.3.1. Pattern altitude is 2,000 feet MSL (1,200 feet AGL). Weather criteria: ceiling at or above 1,700 feet AGL and visibility at or above 5,000 m.

3.2.3.2. Pattern Profile. Aircraft must request and tower must specifically approve “inside” or “outside” closed traffic prior to the aircraft commencing the procedure. See **Attachment 8**, Class D Airspace/VFR Overhead/Inside/Outside Closed Pattern.

3.2.3.2.1. Outside closed traffic pattern profile. Remain outside Ramstein Village, Miesenbach, and Ramstein AB housing area, but inside Mackenbach.

3.2.3.2.2. Inside closed traffic pattern profile (Runway 26/27 only). Remain inside Ramstein AB, avoiding Ramstein Village, Ramstein High School, and USAFE/NATO Headquarters buildings to the maximum extent possible.

3.2.4. Overhead Pattern.

3.2.4.1. Pattern altitude is 2,500 MSL (1,700 ft AGL). Only right breaks will be made to both runways and pilots will remain clear of populated areas listed in **Table A8.2**. See **Attachment 8**, Class D Airspace/VFR Overhead/Inside/Outside Closed Pattern.

3.2.4.2. Weather criteria: ceiling at or above 2,200 feet AGL and visibility at or above 5,000 m.

3.2.4.3. Aircraft entering the overhead pattern from WISKY or ECHO will initially be at 3,000 feet MSL (2,200 ft AGL) then descend to the overhead pattern altitude after passing points WISKY or ECHO. The weather criteria is ceiling at or above 2,700 ft AGL and visibility at or above 5,000 m.

3.2.4.4. Pattern profile from WISKY or ECHO. Proceed to points WISKY at 3,000 feet MSL (2,200 feet AGL) for initial to Runway 08/09 or ECHO at 3,000 feet MSL for initial to Runway 26/27, or as directed by tower. Avoid overflight of Landstuhl/LRMC, south of Bahnstrasse, to the max extent practical. Exceptions are with strong winds requiring pattern correction. Only right breaks will be made to both runways. Exception: If ATC requires C- 21 aircraft to break past the approach end when Runway 08/09 is in use, a left break will be used to avoid overflying Kindsbach village. Additionally, when a left break is used, C- 21 aircraft will avoid overflying Ramstein village to the north and turn base around the water tower.

3.2.5. Protection of the overhead pattern. When the overhead pattern is in use, all departure aircraft will maintain at or below 2,000 feet MSL (1,200 feet AGL) until departure end of runway.

3.2.6. Radar Vectors to Initial. Pilots requesting vectors to initial will be vectored by Ramstein GCA to a point no closer than 7 NM on final approach. The IFR flight plan is cancelled when the aircraft reaches the “initial point” on the initial approach portion of the maneuver. Pilots who do not report the field in sight by 7 NM final will be instructed to climb to at least 4,000 feet MSL, then be radar vectored into the radar pattern for an instrument approach.

3.2.7. Holding procedures. When traffic dictates, aircraft will be instructed to hold at either points WISKY, ECHO, or PIVOT by tower until resumption of the normal traffic pattern can continue. Aircraft are expected to circle to the north at 3,000 feet MSL unless specifically instructed otherwise by tower personnel.

3.2.8. Aircraft will squawk 0033 in the VFR pattern or as directed by ATC.

3.3. VFR Tactical Procedures. See Attachment 10, VFR Tactical Procedures.

3.3.1. High speed downwind.

3.3.1.1. Pattern altitude is 3,000 feet MSL (2,200 feet AGL) from points PIVOT, ECHO, and WISKY, then descend to 2,000 MSL (1,200 feet AGL) according to profile.

3.3.1.2. Weather criteria: ceiling at or above 2,700 feet AGL and visibility at or above 5,000 m.

3.3.1.3. Pattern profiles.

3.3.1.3.1. North high speed downwind. Only available for Runways 26/27. From point PIVOT, proceed between the towns of Miesenbach and Mackenbach at 3,000 MSL (2,200 feet AGL). On the 45-degree entry leg, descend to pattern altitude of 2,000 MSL (1,200 AGL). From point WISKY, proceed between the towns of Miesenbach and Ramstein at 3,000 MSL (2,200 feet AGL) then descend to pattern altitude of 2,000 MSL (1,200 AGL).

3.3.1.3.2. South high speed downwind. Proceed from point WISKY at 3,000 MSL (2,200 feet AGL) on a suggested heading of 110 for a 45-degree entry to runways 26/27; or from point ECHO at 3,000 MSL (2,200 feet AGL) on a suggested heading of 240 for a 45-degree entry to Runways 08/09, or as directed by tower. After crossing the runway centerline, descend to 2,000 MSL (1,200 AGL) and enter downwind on the south side of the runway. Remain north of Landstuhl and Kindsbach.

3.3.2. Random shallow approaches.

3.3.2.1. Pattern altitude is no lower than 1,300 feet MSL (500 feet AGL). Weather criteria: ceiling is normally 6,000 MSL (5,200 feet AGL) or 5,700 feet AGL or at least 500 feet above requested altitude and visibility at or above 5,000 m.

3.3.2.2. Pattern profile. Request the random shallow “straight-in” approach via points ECHO for Runway 26/27 or WISKY for Runway 08/09. Request the “random shallow abeam” with a 90-degree entry midfield for a left 270-degree pitch to final, authorized for Runway 26/27 only. Enter Class D airspace no lower than 1,300 feet MSL (500 feet AGL), up to 250 Knots Indicated Airspeed (KIAS), and avoid all no-fly areas. Break to the north. Subsequent patterns will reenter via points WISKY or ECHO, avoiding the no-fly areas. Pitch north on Runway 08/09 is not authorized. Prior to a pitch south on Runway 08/09, the aircraft commander should coordinate with tower. Pilots will use the phraseology; “random shallow straight in” or “random shallow pitch north” for Runways 26/27 or “random shallow pitch south” for Runway 08/09. Approach not authorized between sunset and sunrise.

3.3.2.3. During the abeam maneuver, tower may direct a north downwind entry when another aircraft is 2 miles or less from touchdown, to ensure de-confliction in the event of a missed approach.

3.3.3. Random steep approaches.

3.3.3.1. Pattern altitude is normally 5,500 feet MSL (4,700 feet AGL) or 6,000 feet MSL (5,200 feet AGL) (depending on aircraft type), unless coordinated otherwise with ATC.

3.3.3.2. Weather criteria. Ceiling at least 500 feet AGL above requested pattern altitude and visibility at or above 5,000 m.

3.3.3.3. Pattern profile. Contact tower approaching the airfield and request a climb to 5,500 feet MSL (4,700 feet AGL) (C-130), 6,000 feet MSL (5,200 feet AGL) (C-21), or coordinated altitude for the random steep approach. Pilots will give tower their direction and distance when

approaching the field and coordinate the break with tower. Avoid the no-fly areas in the vicinity of Ramstein AB throughout the maneuver, even at altitude.

3.3.4. VFR high altitude tactical straight-in approaches.

3.3.4.1. Pattern altitude as coordinated.

3.3.4.2. Weather criteria. Ceiling at least 500 feet AGL above requested altitude and visibility at or above 5000 m.

3.3.4.3. Pattern profile. Contact Langen ACC, Ramstein GCA, or tower (depending upon tactical situation) approaching the airfield and coordinate for a climb/descent to requested altitude for the high altitude tactical straight-in approach. Pilots will specify their direction and distance when approaching the field. Request "High Altitude Tactical Straight-In" with altitude request (usually 10,000 feet MSL). Descend at no more than 250 KIAS. Maintain 1,300 feet MSL (500 feet AGL) until glide path intercept. Avoid the no-fly areas in the vicinity of Ramstein throughout the maneuver.

3.3.5. VFR low altitude tactical straight-in approaches.

3.3.5.1. Pattern altitude as coordinated.

3.3.5.2. Weather criteria: ceiling at or above 1,500 feet AGL and visibility at or above 5,000 m.

3.3.5.3. Pattern profile. Contact Langen ACC, Ramstein GCA, or tower (depending upon tactical situation) approaching the airfield and coordinate for a climb/descent to requested altitude. Pilots will specify their direction and distance when approaching the field.

3.3.6. VFR Departure Procedures. Unless otherwise directed/approved by the tower, all fixed wing VFR aircraft shall depart Class D airspace as depicted in **Attachment 11**, VFR Departure Pattern, and avoid overflying local villages for noise abatement. See **Attachment 8**, Class D Airspace/VFR Overhead/Inside/Outside Closed Pattern.

3.3.6.1. Fighters departing VFR must climb to 4,000 feet MSL (3,200 feet AGL) or higher for noise abatement purposes. Weather minimums required for fighter VFR departures will be ceiling at or above 3,700 feet AGL and visibility at or above 5,000 m.

3.3.6.2. Spiral-up departures.

3.3.6.2.1. Weather criteria: ceiling must be at least 500 feet AGL above requested altitude and visibility at or above 5,000 m. Aircraft must maintain VMC throughout the maneuver.

3.3.6.2.2. Pattern profile. Request "spiral-up departure" with ground control during engine start/taxi or with tower when in the pattern. Read back controller's authorization. Initiate turns after departure, airspeed permitting, (similar to inside closed). Turn direction is normally to the north. Aircraft will remain with tower, fly a 180/360-degree climbing turn, cross over the field (enter downwind) at the requested altitude (usually 5,500 feet MSL (4,700 feet AGL)), and remain within 4 NM of the field.

3.4. Special Procedures.

3.4.1. Helicopter operations. In addition to the runway, tower may authorize helicopters to arrive/depart from any portion of the CMA or from any active taxiway. Helicopters will not be authorized to land directly on or takeoff directly from parking ramps. Additionally, controllers will not allow helicopters to over-fly Mitchell Ave on takeoff or landing. Inbound helicopters from the

east, landing on Taxiway Delta and destined for Ramp 3 or Ramp 4 shall be instructed to land prior to (East of) Mitchell Ave and taxi through the Mitchell Ave intersection. Helicopters departing toward the East shall be taxied onto Taxiway Delta and through the Mitchell Ave intersection before being issued takeoff clearance.

3.4.1.1. Helicopter procedures are published in the Department of Defense VFR, Arrival/Departure Routes Europe.

3.4.1.2. Helicopters from Landstuhl Regional Medical Center Helipads/Airport (ETIP) are authorized VFR practice approaches at Ramstein AB. Pattern altitudes are at or below 700 feet AGL/1,500 feet MSL.

3.4.1.3. Helicopters operating at Landstuhl Regional Medical Center Helipads/Airport (ETIP) are authorized to operate at or below 100 feet AGL for activities contained on the helipad/airport proper such as hover taxi, maintenance checks, etc., without contacting tower for approval. All departing helicopters must contact tower prior to take-off in order to receive clearance.

3.4.2. Restricted low approaches.

3.4.2.1. Tower is authorized to restrict low approaches at or above 1,300 feet MSL (500 feet AGL) or 1,800 feet MSL (1,000 feet AGL) for heavy aircraft, when vehicles, personnel, and equipment are on the runway.

3.4.2.2. Tower will advise aircraft of personnel on the runway and will advise personnel on the runway of restricted low approach traffic.

3.4.3. Functional Check Flights (FCF). All FCF and maintenance check flight requests must be coordinated through the 86 OG/CC for approval.

3.4.4. Drop Zone (DZ) Procedures/Paradrop Operations. Airdrop operations on the Ramstein DZ are subject to 86 OG/CC approval on a case-by-case basis. Procedures for paradrop operations and dimensions of the DZ are outlined in the Ramstein AB Paradrop Operations Letter. See **Attachment 14** for Drop Zone Depiction. (OPR: AM).

3.4.5. Combat Off-Load Training. Only 86 AW C-130 aircraft may execute combat offload training unless authorized by the 86 OG/CC. All combat offload training must be coordinated and approved by AM at least 24 hours in advance.

3.5. Reduced Same Runway Separation Procedures.

3.5.1. RSRS standards shall be applied IAW AFMAN 13-204v3. RSRS applies to USAF aircraft only unless authorized by letter of agreement IAW USAFE guidance.

3.5.2. RSRS standards may be applied in separation of:

3.5.2.1. C-130 aircraft and other C-130 aircraft.

3.5.2.2. Fighter or attack type aircraft and other fighter or attack type aircraft.

3.5.3. RSRS may be applied to:

3.5.3.1. Full stop behind a full stop, low approach, or touch-and-go.

3.5.3.2. Touch-and-go behind a touch-and-go or low approach (C-130s excluded from RSRS between a touch-and-go and another touch-and-go).

3.5.3.3. Low approach behind a low approach.

3.5.3.4. Low approach behind a full stop, in which case the succeeding aircraft will offset laterally to not over fly the aircraft on the runway. For C-130s that are members of the same formation, the succeeding C-130 must maintain at least 500 feet lateral or vertical separation when over-flying the C-130 on the runway.

3.5.4. RSRS is not authorized when:

3.5.4.1. Either aircraft is an emergency.

3.5.4.2. Either aircraft is a heavy.

3.5.4.3. Either the succeeding or preceding aircraft is cleared for the option or a stop-and- go. Exception: RSRS is authorized when the succeeding aircraft is cleared for an option or stop-and-go behind a low approach.

3.5.4.4. The RCR is less than 12 or breaking action reports of less than "medium" are reported.

3.5.5. The minimum RSRS authorized is:

3.5.5.1. 3,000 feet between same type fighter and attack aircraft (e.g., F-16 behind an F- 16) during daylight hours.

3.5.5.2. 6,000 feet between:

3.5.5.2.1. Anytime reported braking action is less than good.

3.5.5.2.2. Dissimilar fighter and attack aircraft (e.g. F-15 behind an F-16).

3.5.5.2.3. Same type fighter and attack aircraft during nighttime hours.

3.5.5.2.4. A landing (single aircraft or formation) behind a formation landing.

3.5.5.2.5. A formation landing behind a full stop.

3.5.5.2.6. A C-130 and another C-130.

3.5.5.3. Weather must be at or above a 500 feet ceiling and 2,400 meters visibility for nighttime RSRS operations.

3.5.5.4. C-130 formations may perform nighttime RSRS operations using the weather minima prescribed by AFMAN 11-2C-130, Volume 3, C-130 Operations Procedures.

3.6. Intersection Departures. Controllers may omit the remaining distances for locally assigned aircraft.

3.6.1. Runway 08 intersection departures are available from:

3.6.1.1. Taxiway Echo: south – 4,695 feet remaining, north – 4,830 feet remaining.

3.6.1.2. Taxiway Golf, 8,852 feet remaining.

3.6.1.3. Taxiway Lima, 3,846 feet remaining.

3.6.2. Runway 26 intersection departures are available from:

3.6.2.1. Taxiway Bravo, 8,783 feet remaining.

3.6.2.2. Taxiway Echo: south – 4,920 feet remaining, north – 4,758 feet remaining.

3.6.2.3. Taxiway Lima, 5,928 feet remaining.

3.6.3. Runway 09 intersection departures are available from:

3.6.3.1. Taxiway Golf, 7,532 feet remaining.

3.6.3.2. Taxiway Echo: south – 3,788 feet remaining, north – 3,950 feet remaining.

3.6.4. Runway 27 intersection departures are available from:

3.6.4.1. Taxiway Charlie, 9,213 feet remaining.

3.6.4.2. Taxiway Echo: south – 5,295 feet, north – 5,120 feet remaining.

Chapter 4

INSTRUMENT FLIGHT RULES (IFR) PROCEDURES

4.1. Radar Traffic Patterns. The radar traffic pattern will normally be flown at 4,000 or 5,000 feet, north downwind, and pilots can expect vectors from Ramstein GCA to the final approach course. See **Attachment 7**, GCA Radar Pattern.

4.2. Local Departure Procedures. Standard instrument departure procedures are listed in the DoD FLIP (Terminal) Vol-3, High and Low Altitude Europe, North Africa and Middle East.

4.2.1. Expect TOLEY 2 departure for Runway 26/27.

4.2.2. Expect BOLKI 2 departure for Runway 08/09.

4.2.3. Departure climb gradients.

4.2.3.1. Runway 08: Cross DER at least 16’ AGL/792’ MSL. Minimum climb rate 250 ft/NM to 2000’ MSL.

4.2.3.2. Runway 09: Cross DER at least 16’ AGL/796’ MSL. Minimum climb rate 250 ft/NM to 2000’ MSL.

4.2.3.3. Runway 26: Cross DER at least 16’ AGL/776’ MSL. Minimum climb rate 214 ft/NM to 2200’ MSL.

4.2.3.4. Runway 27: Cross DER at least 16’ AGL/776’ MSL. Minimum climb rate 214 ft/NM to 1600’ MSL.

4.3. Standard Climb-out Instructions/Procedures. Transient aircraft must be issued detailed standard climb out instructions as required. Prior coordination with Langen ACC must be accomplished for all aircraft departing GCA airspace. The use of the phrase, “execute local climb-out” applies to locally-assigned aircraft only per standard local climb-out outlined in **Paragraph 4.3.1** below.

4.3.1. Standard local climb-out procedures.

4.3.1.1. ATC Phraseology: “EXECUTE LOCAL CLIMBOUT, (state applicable minimum climb gradient)”

4.3.1.2. Runway 26/27 local climb-out: FLY RUNWAY HEADING, CROSS DEPARTURE END AT OR BELOW 2,000 FEET MSL, CLIMB AND MAINTAIN 4,000.

4.3.1.3. Runway 08/09 local climb-out: FLY RUNWAY HEADING. CROSS DEPARTURE END AT OR BELOW 2,000 FEET MSL, CLIMB AND MAINTAIN 4,000. CLIMB GRADIENT 250 FEET PER NAUTICAL MILE.

4.3.1.4. The GCA shall have control of departures at or beyond three (3) DME unless otherwise coordinated with tower.

4.4. IFR Arrival Procedures.

4.4.1. Unless otherwise requested, pilots should expect ILS approach.

4.4.2. Reference speeds. C-130: 170 KIAS; C-21: 200 KIAS; and C-37: 200 KIAS.

4.4.3. Circling approaches are available upon request, traffic permitting. Circling is not authorized south of Runway 08/26 IAW DoD Terminal FLIP.

4.4.4. Full stop/stop and go operations. Aircraft that desire a full stop/stop and go landing shall advise both GCA and tower on initial call-up to aid in sequencing other aircraft in the pattern.

4.4.5. C-130 formation procedures. GCA should allow for 1 NM per aircraft in formation between turns/vectors (i.e., flight of four requires 4 NM between turns/vectors).

4.5. Breakout/Go-Around/Missed Approach Procedures.

4.5.1. Breakout procedures. When an aircraft/flight is at 6 miles or more from end of runway: "FLY RUNWAY HEADING, CLIMB AND MAINTAIN 5,000" unless otherwise directed by GCA.

4.5.2. Go-Around procedures. When an aircraft/flight is less than 6 miles from the end of the runway, the aircraft will be instructed to execute local climb out, unless otherwise directed by tower. Note: In the event of an aircraft in departure position on the runway, tower will issue "GO AROUND LEFT (Runways 26/27)/RIGHT (Runways 08/09) OF RUNWAY." Aircraft will offset 200 feet south of the runway, and then fly local climb out procedures.

4.5.3. Missed approach procedures. Pilots will fly the published missed approach procedure for the published approach they are executing, unless otherwise instructed by ATC. If no published missed approach procedure exists, pilots will execute local climb out procedures and notify ATC.

4.6. Station Keeping Equipment (SKE) Procedures. C-130J aircraft are equipped with SKE technology which allows pilots to maintain pre-set formation parameters using either manual or automated flight.

4.6.1. ATC considerations for dealing with SKE formations are listed below.

4.6.1.1. Spacing and Airspeed. Formations will establish standard spacing of 6,000 feet in trail from the previous aircraft in preparation for the instrument approach. Formation aircraft will slow from their enroute airspeed to 150 KIAS within 10 miles of the FAF and will slow to 135 KIAS within 1 mile of the FAF. Formation aircraft will maintain 135 KIAS until short final, at which point they will transition to the appropriate approach speed based on individual aircraft weight and wind conditions. Speed restrictions for formation aircraft executing SKE procedures are not authorized.

4.6.1.2. Vectors. Due to SKE technology limitations, GCA vectors of 90 degrees or less are preferred. If a turn of greater than 90 degrees is required, controllers will break the turn into two or more subsequent vectors, if able. In the interest of flight safety, pilots must comply with ATC instruction at all times and cannot resume their own headings without ATC approval. All vectors and headings will be situational based upon traffic.

4.6.1.3. Holding. SKE formations are restricted in their maneuverability when entering a holding pattern. If holding is required as a result of a delay or cancellation of approach clearance, "large box" vectors are preferred. If unable to accommodate this request, expect that the lead aircraft to coordinate for formation split prior to holding pattern entry. Aircraft will continue the approach as separate elements.

4.6.1.4. Altitudes. When traffic and airspace permit, the earliest possible descent from an enroute altitude to a FAF altitude is preferred and will best set up the formation for an instrument approach.

For example, clearing the formation to 5,000 feet no later than 10 NM prior to point XIDOD allows the formation to capture the glideslope prior to the FAF while complying with approach restrictions. Descent will be situational based on traffic.

4.6.2. Standard Local SKE Routes. The following routes are the standard IFR routing for 86 AW C-130J SKE formations. Formation lead calls Ramstein Ground for formation engine start with the requested route. Ground control will contact Langen ACC for the clearance. The following routes are only available for departures from Runway 26/27.

4.6.2.1. Marnheim SKE North. Fly the TOLEY2N departure to RUDOT. Routing following RUDOT is direct OBIGA, direct ABSIX, then Q762 to BOLKI, direct HDM, direct ANEKI, direct MARIP (local custom data point at RMS/106/29), then direct MARDZ (RMS/059/22).

4.6.2.2. Marnheim SKE South. Fly the TOLEY2S routing through PIREK, then direct MARIP, then direct MARDZ.

4.6.2.2.1. Following the simulated airdrop at MARDZ, the formation will request own navigation for the Marnheim SKE recovery to the ILS Runway 26/27 (or as required) from Langen and/or GCA. The routing for the Marnheim SKE recovery is direct REC1 (RMS/039/19), direct REC2 (RMS/026/13), direct REC3 (RMS/045/10), direct REC4 (RMS/069/26), direct REC5 (RMS/085/25), then direct XIDOD.

4.6.2.2.2. When Runway 08/09 is in use, GCA should plan to vector the formation south of the runway for a right base turn to the requested approach.

4.6.2.3. Knight SKE North. Fly the BOLKI2N departure to OLIVI. Routing following OLIVI is direct TUTOV, direct ROPUV, direct RUDOT, direct IDARO, direct TOLEY, then direct MAPIG for the ILS 08/09 for recovery procedures. A simulated airdrop will occur at point IDARO.

4.6.2.4. GTQ SKE. If departing Runway 26/27, fly the TOLEY2S transition to SARRI, then direct point MAKOT to transition to French ATC control. If departing Runway 08/09, fly the BOLKI2W transition to SARRI, then direct point MAKOT to transition to French ATC control. Aircraft will recover via MAKOT Z818 MANEM Q762 XIDOD for Runway 26/27 operations. If recovering to Runway 08/09, proceed MAKOT direct MAPIG.

4.7. Availability Restrictions for DASR Approaches and PAR Approaches/Monitoring. Ramstein AB does not have DASR or PAR approaches available.

Chapter 5

EMERGENCY PROCEDURES

5.1. Operation of Primary Crash Alarm System (PCAS) and Secondary Crash Net (SCN).

5.1.1. The PCAS is comprised of voice equipment designed to provide a direct line of communication between the tower and Fire Alarm Communications Center, Airfield Management, and Medical Services.

5.1.2. Activation. Tower will activate the PCAS for all situations that place an aircraft in danger and actual/simulated disaster situations. Tower will test the PCAS daily between 0800L and 0815L. On weekends and holidays, medical services are handled by the Fire Department. Activation is required for, but not limited to the following situations:

5.1.2.1. In-flight/ground emergencies

5.1.2.2. Hijack/unauthorized aircraft movements

5.1.2.3. Bomb threats

5.1.2.4. Primary/alternate tower evacuation

5.1.2.5. Fuel spills

5.1.2.6. Unscheduled barrier engagements

5.1.2.7. Hung ordnance/hung flare

5.1.2.8. Hot brakes

5.1.2.9. Emergency Power Unit (EPU) activation (actual or suspected)

5.1.2.10. No radio aircraft

5.1.2.11. Pilot’s request

5.1.2.12. Local exercise information (as required)

5.1.2.13. Any other situation deemed necessary by the tower/GCA Watch Supervisors

5.1.3. During wing/base exercises, information passed via the PCAS will be preceded and followed by the words “EXERCISE, EXERCISE, EXERCISE.”

5.1.4. For aircraft emergencies, tower personnel will obtain and pass the following information via the PCAS:

5.1.4.1. Aircraft identification and type.

5.1.4.2. Nature of emergency.

5.1.4.3. Pilot’s intentions.

5.1.4.4. Location of aircraft for ground emergencies.

5.1.5. As able, without delay of the PCAS activation, obtain and relay the following:

5.1.5.1. Fuel remaining in time and pounds. Time information can be used for ATC purposes, pounds should be forwarded for use by fire response personnel.

5.1.5.2. Number of personnel on board.

5.1.5.3. Estimated time of arrival.

5.1.5.4. Type armament.

5.1.5.5. Landing runway.

5.1.5.6. Anticipated approach or departure end barrier engagement.

5.1.5.7. Other information per FAAO 7110.65, as necessary.

5.1.5.8. If available, aircraft tail number for ground emergencies.

5.1.6. SCN. The SCN is comprised of voice equipment designed to transmit information critical to aircraft and airfield operations. These notices may be received from tower or from off-base personnel reporting an inbound emergency or aircraft crash. AM will test the SCN between 0815 and 0900L daily. Note: If information requiring an SCN activation is received from an off-base source, AM will contact the tower, GCA, and/or command post to verify the information prior to activating the SCN.

5.1.7. SCN activation. The SCN will be activated by AM when:

5.1.7.1. The PCAS has been activated by the tower. All information from the PCAS will be relayed verbatim on the SCN.

5.1.7.2. The following conditions exist: weather warnings, IFEs, GEs, change in Force Protection Condition (FPCON) level, Disaster Response Force (DRF) activations/recalls, bomb threats or terrorist activities, or at the request of the Emergency Operations Center (EOC) Director.

5.1.7.3. Agencies requesting activation of the SCN will pass their message on the circuit after all agencies have answered.

5.1.7.4. When AM is manned with only one person and required on the airfield, they will notify 86 AW/CP and tower. 86 AW/CP will assume responsibility for activating the SCN in their absence. AM personnel will return to the office as soon as possible when notified by tower to activate the SCN. If unable to return in time, Tower may call CP and advise CP of the emergency so they may ring out the SCN.

5.1.8. The requirements listed in **Paragraph 5.1.2** for activating the PCAS also apply to the SCN.

5.1.9. SCN agencies:

5.1.9.1. Fire Department

5.1.9.2. 86 AW Command Post

5.1.9.3. AMC Control Center

5.1.9.4. Security Forces

5.1.9.5. MOC

5.1.9.6. Transient Alert

5.1.9.7. Weather

5.1.9.8. 86 OG

5.1.9.9. CE Service Desk

5.1.9.10. Flight Safety

5.1.9.11. Emergency Management

5.1.9.12. EOD

5.1.9.13. Crash Recovery

5.1.9.14. Flight Medicine

5.2. Emergency Response Procedures. In-Flight/Ground Emergency Procedures (On/Off Base).

5.2.1. The senior Fire Department official on duty will be the Emergency incident commander (IC) for all emergencies on the airfield.

5.2.2. The crash/emergency response force will assemble at pre-determined taxiways adjacent to the runway, as required by the type or location of the emergency.

5.2.3. During an emergency, communications on the Ramp Net shall be kept to a minimum.

5.2.4. AM will respond and stage based on the situation. For IFE, AM will stage toward the approach end of the runway. AM will conduct a FOD check after aircraft arrival and when deemed safe by the IC. AM will suspend/close/resume operations as needed.

5.3. External Stores Jettison Area Procedures. Follow the aircraft abandonment procedures in **Paragraph 5.7** and those depicted in **Attachment 12**, Jettison of External Stores and Cargo/Aircraft Abandonment Area.

5.4. Fuel Dumping. Fuel dumping is not permitted within the confines of GCA airspace. The GCA will coordinate with Langen ACC for aircraft requesting fuel dumping. GCA and Tower will report incidents of fuel dumping to the AOF/CC.

5.5. Emergency Aircraft Arresting System Procedures. All information regarding barrier engagement procedures, responsibilities, suspension and resumption of runway operations, etc. can be found in **Paragraph 1.8.4** and **Attachment 3**, Aircraft Arresting Barrier Engagement Procedures.

5.6. Hot Brake Area and Procedures.

5.6.1. Pilots will notify the tower any time an overheated brake condition is suspected. Tower will activate the PCAS and advise all agencies of the ground emergency.

5.6.2. If conditions permit, utilize hot brake areas IAW **Paragraph 1.15.1**.

5.6.3. If hot brakes are detected, the senior fire official/crash recovery crew will advise the aircrew to shut down by giving the pilot the cut engine(s) signal when circumstances require engine shut down.

5.6.4. Aircraft must be isolated a minimum of 30 minutes (or in accordance with applicable MDS Technical Orders) after hot brakes are declared and then verified by crash recovery crews.

5.7. Abandonment of Aircraft (Controlled Bail-Out, Ejection, Plotting Aircraft coordinates).

5.7.1. The designated aircraft abandonment/egress/jettison area is Baumholder Army Artillery Range depicted in **Attachment 12**, Jettison of External Stores and Cargo/Aircraft Abandonment Area. Impact is desired at RMS TACAN R-331/11.5 DME.

5.7.2. GCA will coordinate vectors to Baumholder. GCA will coordinate range entry with Langen ACC via hotline or Baumholder Range Command (see **Attachment 13**, Consolidated Numbers and Frequencies for contact info). Once on range, aircraft are to remain north of the town of Baumholder and egress/jettison on a 097 heading at 3,600 feet MSL (2,000 feet AGL) to impact at N 49-38.5 E 007-24.5 (RMS 331/11.5). If aircraft are unable to obtain a local ATC handoff, clearance may be obtained by free call to Langen ACC. Langen minimum vectoring altitude is 5,000 feet MSL.

5.7.3. Aircraft need to adjust actual ejection/jettison timing accordingly so that impact occurs within 1 NM of the desired impact point.

5.8. Personnel/Crash Locator Beacon Signal/Emergency Locator Transmitter (ELT).

5.8.1. Notifications. Tower will notify AM and GCA (Langen ACC, when GCA is closed) when an unplanned ELT is heard. AM will notify 86 AW/CP, who will then coordinate with 86 MXG/MXOC (MOC), 721 AMXS/MOC, TA, and 86 OSS Aircrew Flight Equipment (86 OSS/OSL) to initiate ramp checks.

5.8.1.1. AM will continue to monitor the status of the ELT until notification is received of the termination time and location. If ELT does not terminate within an hour or is not located during ramp checks, AM will re-accomplish notification.

5.8.2. ELT testing. Scheduled operational testing for ELTs will be conducted within the first 5 minutes of the hour. Duration of the test will be no more than three audio sweeps. Beacon tests are to be conducted in a shielded or screened room.

5.9. Hung Ordnance Procedures.

5.9.1. All aircraft experiencing hung ordnance/hung flare or other ordnance malfunctions will declare an emergency with live/heavyweight “hung ordnance” as soon as radio contact is established with GCA or tower.

5.9.2. Tower will activate the PCAS. 86 AW/CP will dispatch 786 CES/CED (EOD). EOD, TA, and Fire Department will respond.

5.9.3. Pilots shall fly a straight-in pattern to the maximum extent possible in order to avoid populated areas, factors permitting (e.g., weather, winds, flight safety considerations, etc.). After landing, the aircraft will be taxied to the respective de-arming area IAW **Paragraph 1.15.1**. Note: Hung BDU-33/MK-106 training ordnance requires an emergency response.

5.10. Wind Limitations on Control Tower.

5.10.1. The tower structural wind limitation is 82 knots.

5.10.2. The tower maximum safe wind velocity is 60 knots.

5.10.3. Tower controllers shall follow evacuation procedures IAW Tower Evacuation Checklist when the wind velocity is sustained and/or gusting 60 knots and request a NOTAM be dispatched.

5.11. Evacuation of Airfield Operations Facilities and Alternate Facility Procedures.

5.11.1. Control Tower Evacuation. Tower personnel will evacuate to the alternate tower, Bldg. 2306, located north of Runway 09/27 midfield.

5.11.2. Alternate tower operations are conducted the last Friday of each month from 1300L to 1500L, unless otherwise directed by the tower Chief Controller (CCTLR). A NOTAM will be published if operations will not take place during the prescribed time periods.

5.11.3. In the event of an evacuation to the alternate tower, the following procedures and restrictions apply:

5.11.3.1. The alternate tower visibility on the ground and in the pattern is limited, due to the position and height of the alternate facility. Vehicles and taxiing aircraft shall use caution during operations.

5.11.3.1.1. Ramps 1, 2, 3, 4, and 7 and Taxiways Charlie, Delta, and the northern portion abeam Ramp 4 of Echo are not visible from the alternate tower.

5.11.3.1.2. The VFR patterns will be closed for pattern work.

5.11.3.2. All aircraft shall be transferred to GCA control or Langen ACC during GCA closures until the alternate tower is operational, normally within 20 minutes of initial evacuation time. GCA will monitor 133.205/284.675 during evacuation to the alternate tower.

5.11.3.3. Tower will broadcast notifications to aircraft and vehicles of evacuation and instructions on all assigned frequencies, including 121.5, 243.0, and the Ramp Net, time permitting.

5.11.3.4. All arriving helicopters will be cleared to land or depart via the active runway. Except for Taxiways Delta and Foxtrot, tower Watch Supervisors may authorize helicopters to depart from any active taxiway when an operational advantage can be gained.

5.11.3.5. AM will initiate NOTAM actions as necessary.

5.11.3.6. The tower shall notify AM and GCA when the facility is evacuating to the alternate tower.

5.11.3.7. The vehicular traffic signal controls are unavailable in the alternate tower and will not be activated by tower. Personnel transiting these routes must use caution.

5.11.4. GCA Evacuation. GCA personnel will evacuate to Bldg. 2303, or as determined by GCA CCTLR. In the event of an evacuation of the GCA, the following procedures apply:

5.11.4.1. GCA will transfer aircraft control to Langen ACC and follow applicable checklists for personnel evacuation.

5.11.4.2. Tower will monitor 124.28 and 284.675 and instruct all aircraft to contact Langen ACC. Note: Tower cannot monitor GCA frequencies when operating in the alternate tower facility.

5.11.4.3. The GCA shall notify AM and tower when the facility is evacuating. 5.11.4.4. AM will initiate NOTAM actions as necessary.

5.11.5. AM Evacuation. AM personnel will evacuate to Bldg. 2318. AM dispatchers shall follow evacuation procedures IAW the AM evacuation checklist.

5.11.5.1. AM will notify tower, GCA, AOF, and activate the SCN.

5.11.5.2. Upon resumption of normal operations, AM will notify the agencies in **Paragraph 5.11.5.1** and again activate the SCN.

5.12. Other Emergency Procedures.

5.12.1. Aircraft Mishap and Hazardous Air Traffic Report (HATR) Response Procedures.

5.12.1.1. In the event of an aircraft mishap, the tower will activate the PCAS and ATC/AM will initiate Mishap/HATR checklists.

5.12.1.2. The senior Fire Department official will be the IC unless otherwise delegated.

5.12.1.3. ATC and AM will relay all information to the IC and provide assistance to the maximum extent possible. If necessary, AM will suspend runway operations.

5.12.1.4. For further information, contact 86 AW/XP.

Chapter 6

FLIGHT PLANNING PROCEDURES

6.1. Responsibilities.

6.1.1. Aircraft commanders:

6.1.1.1. Ensure IFR flight plans are filed at least four hours prior to estimated time of departure (ETD) and VFR flight plans are filed one hour prior to ETD. Special operations and procedures may require earlier filing. Consult AP/2 for additional lead times.

6.1.1.2. All flights departing ETAR must have a flight plan on file with AM Exceptions are 618 AOC aircrews with a flight plan on file with MAJCOM flight planning cell, contracted air carriers, general aviation, Army helicopters who have flight plans on file with Army Flight Operations Detachment, or locally assigned units whose main flight planning program (Fore flight, GDSS2) allows users to retrieve flight plans for flight following purposes and maintains a copy IAW Air Force RDS, Table 13-06, Rule 4.00.

6.1.1.2.1. Use DD Form 1801, DoD International Flight Plan IAW AFI 11-202v3, General Flight Rules and FLIP General Planning.

6.1.1.2.2. Use DD Form 1801 or contact dispatch agency for all changes to flight plans prior to departing.

6.1.2. AMC procedures.

6.1.2.1. AMC aircrews will have flight plans pre-filed through the integrated flight manager.

6.1.2.2. AMC aircrews or AMCC will coordinate any flight plan changes with the integrated flight manager.

6.1.2.3. Flight plans filed electronically shall be archived in GDSS IAW Air Force RDS, Table 13-07 Rule 3.00.

6.1.3. Locally-assigned aircrews:

6.1.3.1. File flight plans IAW AFI 11-255v3 and local directives.

6.1.3.2. Contact AM to push/activate requested flight plan through GDSS2.

6.1.3.3. Contact local squadron or flight planning facility with the original signed copy of the DD Form 1801 for any flight plan changes needed while en-route.

6.1.4. Transient aircrews must file all flight plans at AM counter. If flight plan was filed at a previous location, AM personnel will determine existence of flight plan in GDSS. If it cannot be located, transient aircrews must re-file.

6.1.5. AM will:

6.1.5.1. Review flight plans filed at AM for compliance with FLIP General Planning.

6.1.5.2. Submit all flight plans for US and foreign aircrews not supported by 603 AOC/A34 or AMC IAW FLIP General Planning.

6.1.5.3. Maintain original flight plans IAW Air Force RDS, Table 13-07, Rule 3.00.

6.1.6. The pilot in command is responsible to identify flight plan changes due to weather, ground aborts, or other mission requirements as soon as possible.

6.1.7. IAW the German Aeronautical Information Publication (AIP), pilots must request a delay message through their flight planners if one of the following situations exists:

6.1.7.1. IFR flight within the Integrated Initial Flight Plan Processing System Zone (IFPZ) if delayed by more than 15 minutes but not more than 20 hours compared with the off- block time.

6.1.7.2. IFR flight outside the IFPZ if delayed by more than 30 minutes compared with the off-block time.

6.1.7.3. VFR flight if delayed by more than 30 minutes compared to the off-block time. Note: Countries in the IFPZ are listed here: **http://www.fsint.de/en/22**.

6.1.8. Problems with flight plans (e.g., route of flight, missing data) and air traffic flow management messaging will be handled by calling the affected flying squadron and coordinating with the on-duty operations officer to correct the problem. The on-duty operations officer shall be responsible for correcting or re-filing the flight plan for the aircrew to prevent mission delay.

6.1.9. Electronic flight plan filing:

6.1.9.1. Configure computers by downloading and following instructions posted on the DD Form 1801.

6.1.9.2. Electronic flight plan and air traffic flow management messaging: Check FLIP AP/2 and download instructions from the Form.

6.2. Slot Times and Flow Control.

6.2.1. The two terms, slot time and flow control, are flight restrictions imposed and controlled by several European air traffic control organizations.

6.2.1.1. Flow control means that at a certain point or region (e.g., flight information region boundary) ATC authorities can and/or will only handle a certain number of aircraft per hour.

6.2.1.2. Slot time, or calculated takeoff time (CTOT), is a 15-minute window during which an aircraft can depart. Slot time is issued to Ramstein from Eurocontrol through GDSS, and then passed to aircraft by tower.

6.2.1.3. To comply with European flow control restrictions, tower controllers will not issue clearances allowing aircraft to depart more than 5 minutes before or 10 minutes after an assigned CTOT.

6.2.2. Flow control exempt flights. Flow control exemption shall be in accordance with FLIP AP/2, Theater Supplementary Notices and Procedures.

6.2.3. VFR Flight Planning Procedures. All VFR flight plans must be filed IAW FLIP General Planning, Chapter 4.

6.3. Ready Messages.

6.3.1. A request for a Ready Message to be sent through GDSS does not guarantee that an aircraft's slot time will be pushed up.

6.3.2. Requests for ready messages can be relayed to AM via the Pilot-to-Dispatch frequency, see **Table 1.1**.

Chapter 7

MISCELLANEOUS PROCEDURES

7.1. Airfield Operations Board.

7.1.1. The Ramstein AB Airfield Operations Board (AOB) will convene quarterly IAW AFMAN 13-204v3. Due to size and complexity of Ramstein AB flying operations, the 86 AW/CV has delegated 86 OG/CC as chairman of the AOB. AOB membership is listed in **Table 7.1**, although other interested agencies may attend.

Table 7.1. Airfield Operations Board Membership.

86 OG/CC (Chair)	86 CEG/CC
37 AS Representative	76 AS Representative
86 OG/OGV	86 AW/SE
86 OSS/CC	86 OSS/OSA
86 OSS/OSW	86 AW/CP
86 CS Representative	86/786 CES/CC
721 AMOG/CC or DO	721 MSS/CC or DO
AMCC Representative	721 APS/CC or DO
86 AW/HN	USAFE TERPS

7.1.2. The following items are required to be briefed at the AOB annually IAW AFMAN 13-204v3, and will be briefed at the AOBs as annotated below:

7.1.2.1. 1st Quarter AOB

7.1.2.1.1. Letters of Procedure: Airfield Operation Instruction, Letters/Memorandums of Agreement, Operations Letters, Operations Plans (OPLANs), Host Nation Agreements.

7.1.2.1.2. Special Interest Items (SIIs)

7.1.2.1.3. Annual Airfield Certification/Safety Inspection and Quarterly Joint Inspection results

7.1.2.2. 2nd Quarter AOB

7.1.2.2.1. Annual Self Inspection Results

7.1.2.3. 3rd Quarter AOB

7.1.2.3.1. Aircraft Parking Plan

7.1.2.4. 4th Quarter AOB

7.1.2.4.1. Terminal Instrument Procedures (TERPS)

7.1.2.4.2. Status of existing airfield waivers (with emphasis on temporary waivers and associated correction plans).

7.2. NOTAM PROCEDURES.

7.2.1. AM is the NOTAM issuing office at Ramstein AB. Agencies requesting NOTAM issuance should contact AM.

7.2.1.1. AM issues Local NOTAMs (L-Series) via the NOTAM Manager website.

7.2.1.2. AM issues Safety NOTAMs (M-Series) via GDSS, which are converted to (P- Series) through the International NOTAM Office (INO).

7.2.2. Tower is the NOTAM monitoring facility for Ramstein AB.

7.3. Flight Information Publication (FLIP) Accounts and Procedures for Requesting Changes.

7.3.1. Procedural changes. HQ USAFE Air Procedures Flight (APF) is responsible for procedural changes and FLIP cycle reviews. Forward procedural change requests to HQ USAFE APF.

7.3.2. Non-procedural changes. Forward non-procedural change requests to the Airfield Manager.

7.3.3. AM is responsible for ordering and maintaining FLIP for the flight planning room and for ATC/AM use. FLIP located in AM and the flight planning room are for transient aircrews only, and are not to be removed from the facility. IAW AFI 11-201, local flying units are responsible for the establishment and management of their own FLIP accounts.

7.4. Prior Permission Required (PPR) Procedures. 618 AOC (TACC) controlled missions must coordinate with 721 MSS for request. 721 MSS will coordinate approval for inbound aircraft the night prior and AM will issue PPR numbers. Request PPRs no earlier than 7 days prior to arrival but no later than 24 hours prior to arrival. IAW AFMAN 13-204v3 USAFE Supplement; AFI 10-1001, Civil Aircraft Landing Permits; AFI 10-1801, Foreign Governmental Aircraft Landings at United States Air Force Installations and this instruction, the Airfield Manager is the delegated approval authority for landing approvals and PPR authorizations. NATO aircraft use of Ramstein AB is authorized by the 86 AW/CC IAW the USAFE-AFAFRICA/CC delegation memo.

7.5. Air Evacuation Notification and Response Procedures. The Fire Department is designated as the agency responsible for rescue protection (aeromedical evacuation flights) upon pilot notification.

7.5.1. AM will notify the Fire Department of inbound aeromedical aircraft landing at Ramstein AB.

7.5.2. Tower will relay any additional information requested by the pilot to the appropriate agency.

7.6. Unscheduled/Unauthorized Aircraft Arrivals. Other than cases of IFE or weather divert, aircraft that are not scheduled to arrive at Ramstein AB will not receive landing clearance from tower. Aeromedical Evacuation (AIREVAC), Special Air Missions (SAM), and Hospital (HOSP) aircraft are exempt from OBO/PPR restrictions but are required to obtain a PPR number when able for tracking and notification purposes. Any uncertainty regarding authorization to land on Ramstein AB must be resolved with AM prior to requesting landing clearance.

7.6.1. In the event of an unauthorized landing, tower activates the PCAS and AM will activate the SCN and coordinate with SFS to respond. The aircraft will be held at a location deemed appropriate by the tower and the on-scene commander until the purpose for the landing is determined. If

applicable, AM will coordinate for Civil Aircraft Landing Permit (CALP) IAW AFI 10-1001 or Aircraft Landing Authorization Number (ALAN).

7.7. Distinguished Visitor Notification Procedures.

7.7.1. Upon notification of a DV flight inbound, AM will add the flight to the PPR log. AM will contact 86 AW/Protocol via phone with inbound information and changes.

7.7.2. GCA (tower during GCA closures) will notify the 86 AW/CP when DV-6 or higher is 25 miles from touchdown or as soon as possible after initial contact if closer than 25 miles.

7.7.3. For all other applicable DV notification procedures, consult AP/2.

7.8. Dangerous/Hazardous Cargo.

7.8.1. The HCP on Ramp 8 is identified as the dangerous/hazardous cargo area and is managed by Cape Forecasting. Ramp 5 and Taxiway Kilo can also be used to park aircraft carrying certain amounts of hot cargo. Servicing (up or downloading) of munitions loads may occur on Ramp 5 but must be limited to the amounts determined by Wing Safety If the amount of munitions exceeds authorized limits, aircraft must be moved to the HCP to service munitions. Taxiway Kilo use for hazardous cargo parking requires an airfield contingency waiver signed by 86 AW/CC.

7.8.1.1. No more than three aircraft can be on the HCP at one time.

7.8.2. If the pilot of an explosives-loaded aircraft or a hazardous cargo aircraft declares an IFE and requires landing at Ramstein AB, the following actions shall be accomplished.

7.8.2.1. Tower shall activate PCAS and relay all information received from the pilot.

7.8.2.2. Upon landing, aircraft will be directed to the HCP and marshalled into position.

7.8.2.3. If the AMCC or 86 AW/CP receives notification (by means other than the PCAS) of an in-flight emergency involving an explosives-loaded aircraft or a cargo aircraft with hazardous cargo aboard, they shall relay all information to tower. Tower will then activate the PCAS.

7.8.2.4. The 86 AW/CP is the designated focal point for information and facilitates coordination of support efforts concerning hazardous munitions/cargo.

7.8.3. Explosive loaded aircraft cargo operations. Refer questions about approved explosive loaded aircraft cargo operations locations/limitations at Ramstein AB to Wing Safety.

7.9. Night Vision Device (NVD) Operations. NVD operations will be conducted IAW established local procedures and with units with an approved LOP with the 86 OG. Tower and AM do not utilize NVDs during operations and NVD operations will not be conducted during alternate tower operations. For airfield driving related NVD operations procedures, reference AFI 13-213 Ramstein Supplement.

7.10. Local Aircraft Priorities. ATC services are provided on a first-come, first-serve basis as circumstances permit, with the exception of the operational priorities listed in FAAO 7110.65. Local aircraft priorities are listed below in the following order:

7.10.1. Emergency aircraft

7.10.2. “LIFEGUARD”; or “AIR EVAC or MED EVAC” (only when specifically requested)

7.10.3. Search and rescue aircraft performing a search and rescue mission

7.10.4. Alert aircraft

7.10.5. Presidential aircraft and support aircraft

7.10.6. Flight Check aircraft

7.10.7. Primary Nuclear Airlift Force/National Airborne Operations Center (NAOC)

7.10.8. Aircraft with a validated controlled departure time (slot time)

7.10.9. DV aircraft

7.10.10. 86 AW formation aircraft

7.10.11. Full stops

7.10.12. AMC Departures

7.10.13. 86 AW practice approaches

7.10.14. All other aircraft on a first-come, first-served basis

7.11. Lost Communication Instructions.

7.11.1. If cleared for an approach, squawk 7600 and execute approach as published and look for light gun signals from tower.

7.11.2. If not cleared for an approach, do one of the following:

7.11.2.1. VMC: FLY DOWN THE RUNWAY AT 2000 MSL ROCKING WINGS AND SQUAWKING 7600, TURN NORTH TO ENTER THE VFR PATTERN AND LOOK FOR LIGHT GUN SIGNALS. Note: When alternate tower operations are in effect, aircraft will not be able to observe light signals until turning final approach.

7.11.2.2. IMC: IF NOT ESTABLISHED ON A PUBLISHED APPROACH, MAINTAIN AT LEAST 5000 AND PROCEED DIRECT XIDOD/MAPIG (DEPENDENT UPON RUNWAY IN USE). IF ALIGNED FOR AN APPROACH, FLY APPROACH AS PUBLISHED. OTHERWISE, EXECUTE ONE TURN IN HOLDING FOR ALIGNMENT, AND FLY APPROACH AS PUBLISHED. SQUAWK 7600. MONITOR TOWER FOR LIGHT GUN SIGNALS. Note: When the alternate tower is in use, aircraft will not be able to observe light signals until approximately 4 NM final.

7.12. Opposite Direction Operations (ODO).

7.12.1. ODOs shall be approved when delays will not be experienced by aircraft utilizing the runway in use.

7.12.2. The tower is the approval authority for opposite direction arrivals and the GCA is the approval authority for opposite direction departures. The facility receiving the opposite direction requests will inform the adjacent facility immediately upon receipt. Both tower and GCA have the authority to deny opposite direction operations based on a workload-permitting basis.

7.12.3. All coordination for ODO must be on a recorded line and IAW AFMAN 13-204v3. Controllers must conduct all coordination on a recorded line. The controller's initial coordination must include, "OPPOSITE DIRECTION DEPARTURE/ARRIVAL, (CALL SIGN) (TYPE AIRCRAFT), RUNWAY (number)." All coordination thereafter between controllers must include the phrase "OPPOSITE DIRECTION DEPARTURE/ARRIVAL, RUNWAY (number)."

7.12.4. Opposite Direction Cutoffs (VFR or IFR).

7.12.4.1. Arrival vs. Arrival (VFR or IFR): First aircraft must have touched down on the runway prior to an opposite direction arrival reaching 10 mile final.

7.12.4.2. Arrival vs. Departure (VFR or IFR): The departing aircraft must be airborne and turned to avoid conflict prior to the opposite direction arrival reaching 10-mile final.

7.12.5. Visual separation is not authorized between two IFR aircraft conducting ODO.

7.12.6. Issue traffic advisories to participating opposite direction aircraft. Example: "OPPOSITE DIRECTION TRAFFIC (distance) MILE FINAL, (type aircraft) or OPPOSITE DIRECTION TRAFFIC DEPARTING RUNWAY (number), (type aircraft). OPPOSITE DIRECTION TRAFFIC, (position), (type aircraft)."

7.12.7. Tower and GCA shall ensure that VFR aircraft conducting ODO are issued a turn to avoid conflict with opposing IFR/VFR traffic.

7.13. Civilian Aircraft Operations. Civil aircraft (except AMC charter), other than emergencies, will not be allowed to land unless an approved CALP/aircraft landing authorization number (ALAN) is on file and verified at AM. Upon verification from AM, a PPR number will be issued to allow for landing/parking at Ramstein AB.

7.14. Civil Use of Military RAWS. Civil aircraft without PPR approval may use USAF NAVAIDs and radar services, as long as operations will not conflict with local traffic.

7.15. Aero Club Operations. Ramstein AB does not have an Aero Club.

7.16. Weather Dissemination and Coordination Procedures: Hazardous/Severe Weather Notification Procedures; Lightning Response.

7.16.1. AM personnel are responsible for disseminating hazardous/severe weather and lightning information via the SCN, IAW AFMAN 13-204v3, Airfield Operations Procedures and Programs and RABI 15-101, Weather Support. Upon receipt or cancelation of a Weather Warning, AM will activate the SCN and broadcast the information over the Ramp Net. AM will publish NOTAMs as necessary.

7.16.2. Lightning watch. This watch is issued 30 minutes prior to thunderstorms occurring within 5 NM radius of the base. Operations or activities may continue; however, all personnel must be prepared to implement lightning warning procedures.

7.16.3. Lightning warning. This warning is in effect whenever lightning is occurring within a 5 NM radius of the base.

7.16.3.1. Personnel in affected locations cease all outside activities and seek shelter. Approved shelters include:

7.16.3.1.1. Dwellings or other buildings protected against lightning.

7.16.3.1.2. Protected underground shelters.

7.16.3.1.3. Large metal-framed buildings.

7.16.3.1.4. Enclosed automobiles, buses, aircraft, and other vehicles with metal tops and bodies.

7.16.3.2. Aircraft with engines running may continue if their specific mission design series regulations allow taxi and takeoff, avoiding all thunderstorms, down drafts or rain shafts. This is

with ATC concurrence and slot time approval. Aircraft may also wait in parking with crew and passengers aboard (this applies to arriving aircraft as well). Aerospace Ground Equipment will be shut down during lightning warnings.

7.17. Airfield Snow Removal Operations. Tower will broadcast an advisory on the ATIS stating: "DE-ICING MATERIAL ON RUNWAY." The advisory will remain on the ATIS for 3 hours after application, or until AM advises de-icing material has dissipated. Tower will advise departing turbojet aircraft of the possibility of smoke on takeoff when chemicals have been applied to the runway. See RABI 32-1002, Snow and Ice Control for more information on snow removal operations.

7.17.1. Snow removal vehicles have right away on the airfield during snow removal operations except for aircraft traffic, or emergency vehicles actively responding to an emergency. All vehicles will ensure they are safe distance away from snow being actively removed from airfield.

7.18. Bird/Wildlife Control: Bird/Wildlife Control : Bird/Aircraft Strike Hazard (BASH) Program Guidelines/Bird Watch Conditions (BWC). The BASH Program is managed by 86 AW/SE. Reference 86 AW OPLAN 91-212, Bird/Wildlife Aircraft Strike Hazard (BASH) Reduction Plan, for 86 OSS/OSA roles and responsibilities.

7.18.1. Bird Watch Condition (BWC) Restrictions to Flying Operations.

7.18.1.1. BWC LOW Restrictions: No restrictions during this condition.

7.18.1.2. BWC MODERATE Restrictions: 86 AW training flights must be approved by the 86 OG/CC or designated representative. Pilots will contact their appropriate squadron for approval routing. This condition requires increased vigilance by all agencies and supervisors and caution by aircrews.

7.18.1.3. BWC SEVERE Restrictions: All arrivals and departures will be delayed until the bird condition is reduced. Minimum fuel or emergency aircraft will be permitted to conduct straight-in, full stop landings only, and will be at the discretion of the pilot-in-command. Departure of any aircraft must be approved by the 86 OG/CC or designated representative. Supervisors and aircrews must thoroughly evaluate mission need before conducting operations in areas under condition SEVERE.

7.19. Supervisor of Flying (SOF) Operating in the Tower. Ramstein AB does not have a SOF program.

7.20. Airfield Photography. Consult 86 AW/PA (Public Affairs) and 86 AW Integrated Defense Plan (IDP) 31-101 for directives regarding photography on or about the airfield. AM is permitted to take pictures on the airfield for construction project and aircraft operation planning.

7.21. Tactical Arrival/Departure Procedures. See **Paragraph 3.2.9** and **Attachment 10**, VFR Tactical Procedures.

7.22. UAS Operations Procedures. Group 1 and 2 UAS may operate in Ramstein Class Delta airspace in and over installation boundaries according to the 86 AW CONEMP for UAS operations and local UAS operating instructions. In absence of these additional operating instructions, the following guidance applies. Whether the UAS flight is planned on-base or off-base, operators will make their request to use Ramstein Class Delta airspace seven days prior to the proposed flight. UAS/RPA capabilities and limitations are defined in AFMAN 13-204v3 and AFI 31-101 USAFE

Supplement. The following guidance applies to UAS/RPA operations for official use purposes only.

7.22.1. UAS Operations Outside Ramstein AB Installation Boundaries. IAW Luftverkehrs-Ordnung (LuftVO [German Air Traffic Regulation]), drone operation within 1.5 km of a military installation must first be granted approval by the aviation authority, which is the 86 AW Installation Program Managers office (IPM). Outside of 1.5 km, IPM must be notified and coordinated with prior to use.

7.22.2. UAS Operations Within Ramstein AB Installation Boundaries. The UAS operator must be trained and qualified to operate UAS according to AFMAN11-502 and shall conduct communications checks with the Tower prior to operations start time. The operator shall request permission from the IPM to operate within the lateral confines of the installation and shall have lost link or line of sight procedures established upon request. IPM will conduct subsequent coordination with 86 SFS, 86 OSS, and 86 CS for force protection, airfield operations, and frequency management concerns. On-base UAS flights are subject to the following restrictions:

7.22.2.1. UAS operations are prohibited during alternate tower operations.

7.22.2.2. Recreational and hobby UAS are NOT authorized on Ramstein AB.

7.22.2.3. Tower may deny an IPM-approved UAS flight due to manned air traffic flow, controller workload, or unsuitable weather conditions.

7.22.2.4. UAS operations beyond visual line of sight of the operator or visual observer are prohibited.

7.22.3. The UAS operator is required to provide IPM the following information by email to **86AW.sUAS.OSO@us.af.mil**.

7.22.3.1. Detailed geographic location and operating area dimensions (e.g., street address, building number, crash grid). Expect delays to approve flights over taxiways and ramps and within 1000 feet of active runways.

7.22.3.2. Highest requested altitude at which operations will be conducted. Expect delays to approve flights at altitudes higher than 400 feet (121 meters) AGL.

7.22.3.3. UAS make and model. Include command and sensor frequencies.

7.22.3.4. Requested date, time, and duration of operations. During adverse weather conditions such as high winds and reduced visibility operations will be denied.

7.22.3.5. Primary and secondary contact information for communications with ATC and method of coordinating initiation and termination of flight operations.

7.22.3.6. Details of purpose of UAS flight request, including the use of audio and/or visual recording devices.

7.22.3.7. Lost link procedures.

7.22.4. If lost link/line of sight of UAS occurs, the operator will immediately notify Tower and provide the following information. If applicable, see **Attachment 13**, Consolidated Numbers and Frequencies, for Ramstein AM’s front desk number.

7.22.4.1. Time of lost link/line of sight.

7.22.4.2. Last known position.

7.22.4.3. Altitude.

7.22.4.4. Direction of flight.

7.22.4.5. Confirm execution of lost link or line of sight procedures.

7.22.4.6. Confirm pilot/observer has visual contact with UAS/RPA.

7.22.5. The safety of manned aircraft will take precedence over unmanned aircraft in all situations.

7.23. Exercise and Real-World Procedures.

7.23.1. Exercise coordination. Base officials must coordinate with the Airfield Operations Flight Commander (AOF/CC) and Airfield Manager at least 48 hours in advance of exercises that involve any ATC facility or the airport movement area. The AOF/CC, Airfield Manager, Tower CCTLR, and GCA CCTLR are considered trusted agents.

7.23.2. During exercises, all participating aircraft will be advised by Ramstein GCA or tower of the alarm condition upon initial contact. During real world events, all aircraft will be advised by Ramstein GCA or tower of the alarm condition upon initial contact. When advised of condition alarm red or black, aircraft will state their intentions in accordance with applicable war plans and simulations. Exercise aircraft will depart and recover IFR, unless otherwise approved by the 86 OG/CC or designated representative.

7.23.3. IAW KMC AF Integrated Defense Plan 31-101 (IDP 31-101), during exercise and real world FPCON Charlie and Delta measures, the tower will ensure this information is contained on the ATIS broadcast.

7.24. Explosive Ordnance Disposal (EOD) Proficiency Range.

7.24.1. EOD operations occur in the SW area of the airfield near Bldg. 2271.

7.24.2. Tower will be notified via landline prior to the start of any explosive operation, and the following information will be passed: number of detonations, time period of explosive operations, and if smoke grenades are being utilized.

7.24.3. Non-emergency range operations require EOD to contact tower via the LMR on the Ramp Net to declare the range is “HOT” prior to beginning explosive operations. EOD will then contact tower via LMR on the Ramp Net at the conclusion of explosive operations to declare the range is “COLD”. Individual detonation notifications are not required unless it is an emergency disposal or involves smoke grenades.

7.24.4. Emergency disposal and smoke grenade operations require EOD to contact tower via the LMR on the Ramp Net to request vertical clearance, inform of number of detonations and time occurring. EOD will proceed with the disposal only after vertical air clearance is approved and notify the tower when detonation has been completed. During all operations, the range safety officer and/or team leader will monitor Ramp Net in case any issues arise.

7.25. Waivers to Airfield/Airspace Criteria.

7.25.1. The airfield manager maintains a copy of all permanent, temporary, and construction airfield waiver packages. The original airfield waivers are located and maintained at 86 CES/CEAOP.

7.25.2. IAW USAFEI 32-1007, the 86 AW/CC is the approval authority for all airfield construction and contingency waivers which impact airfield and/or airspace operations. USAFE/CV is the approval authority for all temporary and permanent waivers to airfield/airspace criteria, except those which affect instrument procedures, and specific airfield markings, see USAFEI 32-1007, Attachment 2 for further details.

7.25.3. Airfield criteria waiver procedures. Airfield Manager, CE, Safety, and 86 OSS TERPS Liaison will:

7.25.3.1. Jointly prepare/initiate waiver requests and coordinate them with HQ USAFE/ APF.

7.25.3.2. Submit requests through the installation to AFIMSC, Detachment 4.

7.25.3.3. Maintain a complete record of all waivers requested and their disposition (approved or disapproved).

7.25.3.4. Refer to Attachment 2 in USAFEI 32-1007, for additional procedures on processing airfield/airspace waivers.

7.26. Wear of Hats and Reflective Belts. All flight line areas are designated no-hat areas. These areas are inside the fence line of the northeast, southeast, and southwest areas, all ramps, taxiways, and end-of-runway areas.

7.26.1. During cold weather, the winter stocking cap, and flyer's helmet (bunny cap) are authorized to be worn in no-hat areas. Remain alert to operating engines.

7.26.2. Security Forces personnel, while performing official duties, may wear the beret with insignia attached; however, when they are within 50 feet of an operating aircraft their berets must be removed and secured.

7.26.3. During inclement weather, darkness or periods of reduced visibility, all personnel working/operating on the flight line are required to wear a reflective belt. Security forces personnel armed to provide security on the flight line are exempt from this requirement.

7.27. Airfield Smoking Policy. IAW AFI 91-203, Air Force Consolidated Occupational Safety Instruction and National Fire Protection Association, National Fire Code 410, smoking is prohibited in aircraft maintenance facilities, flight line areas, and weapons storage and maintenance areas.

7.28. Procedures for Restricting/Closing Airspace for Planned Military Operations.

7.28.1. IAW German Air Traffic Order (LuftVO [Luftverkehrs-Ordnung Ausfertigungsdatum), section 5, § 16, Ramstein AB can request to close airspace to facilitate planned military operations to prevent hazards to the safety of air traffic.

7.28.2. Closure requests must be sent to 86 OSS/OSA NLT one week prior to the requested closure date with the following information: location (latitude/longitude or Ramstein AB's Class D or E airspace), dimensions (radius and altitude), date, and times.

7.28.3. 86 OSS/OSA will send all closure requests to the Federal Ministry of Transport and Digital Infrastructure (BMVI) for approval. If approved, BMVI will issue the appropriate NOTAM.

7.28.4. For unplanned short notice airspace closures, see Langen LOA.

Chapter 8

LOW VISIBILITY PROCEDURES

8.1. Low Visibility Operations.

8.1.1. Tower will notify AM when changes in weather create changes in low visibility conditions. AM will report changes in low visibility conditions via the SCN.

8.1.2. Ramp managers shall establish one central point of contact (POC) for their respective ramps.

8.1.3. When RVR is less than 800 meters and/or the ceiling is less than 200 feet, the CMA is restricted to mission critical vehicles. The following classify as Mission Critical Vehicles: RAWS, Barrier Maintenance, Airfield Lighting, and emergency response vehicles.

8.2. Low Visibility Conditions (VISCONs). When the RVR is at or below 400 meters, four VISCONs exist as defined below.

8.2.1. VISCON 1. VISCON 1 will be in effect when the RVR is between 180 meters and 400 meters, inclusive. While in VISCON 1, all operators on the airfield will wear reflective gear. Only mission critical vehicles are authorized to operate in the CMA and must activate flashing/hazard lights at all times.

8.2.2. VISCON 2. VISCON 2 will be in effect when the RVR is greater than or equal to 100 meters and less than 180 meters. In addition to the provisions listed in **Paragraph 8.2.1**.:

8.2.2.1. Ramp POCs will establish two-way communications with the tower via the Ramp Net.

8.2.2.2. Ramp POCs will take inventory of all personnel and equipment on the ramp.

8.2.2.3. Taxi operations. Only aircraft arrivals will be allowed to taxi.

8.2.2.4. Vehicles will not exceed 10 mph while operating on the airfield.

8.2.3. VISCON 3. VISCON 3 will be in effect when the RVR is greater than or equal to 50 m and less than 100 m. In addition to the provisions listed in Paragraph **8.2.1**–**8.2.2**.:

8.2.3.1. Vehicle operations. Only mission critical vehicles are authorized to operate on the airfield. IAW AFI 91-203, Air Force Consolidated Occupational Safety Instruction, refueling and explosive laden vehicles will not be operated on the airfield.

8.2.3.2. 721 AMXS/MOC and 86 MXG/MOC, will report 100% accountability of personnel, vehicles, and equipment operating on their respective ramp to tower via the landline. Tower will not issue taxi clearances until 721 AMXS/MOC and 86 MXG/MOC report 100% accountability.

8.2.4. VISCON 4. VISCON 4 will be in effect when the RVR is less than 50 meters.

8.2.4.1. Vehicle operations. IAW AFI 91-203, Air Force Consolidated Occupational Safety Instruction, only emergency response and alert vehicles will operate on the airfield. These vehicles will not exceed 10 mph while operating on the airfield. A walking guide equipped with a flashing or luminescent wand is recommended for emergency response or alert vehicles operating on the airfield under these visibility conditions.

8.2.4.2. Vehicles operating on the airfield shall exit all taxiways and await further instructions.

8.2.4.3. Personnel should maintain their current location on the airfield until visibility improves to VISCON 3 or better. Only movement to clear the path for taxiing aircraft or in support of emergencies is authorized.

8.2.4.4. All aircraft engine runs will cease.

8.2.4.5. Taxi operations. Only aircraft arrivals will be allowed to taxi. Prior to an aircraft taxiing to parking, tower will inform each ramp POC in the path of the aircraft for personnel and equipment dispersal. Aircraft will not be given taxi clearance from ATC until each ramp POC in the projected path of the aircraft has ensured and reported the taxi route is free of personnel and equipment.

8.2.4.6. Due to the limitations associated with RVR reporting, 86 OG/CC approval must be obtained to deviate from established procedures. Note: The 86 MXG/CC may approve deviations from procedures listed in this chapter for 86 MXG personnel operating on Ramps 1, 2 and 7; 721 AMXS/CC may approve deviations for 721 AMXS personnel operating on Ramps 5 and 8.

OTIS C. JONES
Commander, 86th Airlift Wing

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

14 CFR Part 91, *General Operating and Flight Rules* 20 September 2023

AFI 10-1001, *Civil Aircraft Landing Permits* 23 August 2018

AFI 10-1801, *Foreign Governmental Aircraft Landings at United States Air Force Installations* 25 September 2018

AFMAN 11-202v3, *Flight Operations* 10 January 2022

AFMAN 11-218, *Aircraft Operations and Movement on the Ground* 24 May 2023

AFMAN 11-230, *Instrument Procedures* 24 July 2019

AFPD13-2, *Air Traffic, Airfield, Airspace and Range Management* 3 January 2019

DAFMAN 13-204_USAFESUP, *Airfield Operations Procedures and Programs* 6 January 2016

AFI 13-207, *Preventing and Resisting Aircraft Piracy (Hijacking) FOUO* 21 June 2010

AFI 13-213_RAMSTEINSUP, *Airfield Driving* 25 April 2016

AFMAN 33-363, *Management of Records* 1 March 2008

AFMAN 91-203, *Air Force Consolidated Occupational Safety Instruction* 25 March 2022

Air Traffic Act Permit 11 June 2003

ETAR - Langen ACC LOA, *Procedures for Provision of Air Traffic Services* 1 September 2023

KMC IDP 31-101, *Integrated Defense Plan* 1 April 2015

Luftverkehrs-Ordnung (LuftVO) Ausfertigungsdatum 29 October 2015

Zentralrichtlinie Militärisches Flugverkehrsmanagement (A2-272/2-2000-14) 30 June 2016

FAAO 7110.65AA, *Air Traffic Control* 5 October 2023

RABI 32-1002, *Snow and Ice Control* 25 May 2019

TO 33-1-23, *Equipment and Procedures for Obtaining Runway Condition Readings* 16 September 2011

UFC 3-260-01, *Airfield and Heliport Planning and Design* 5 May 2020

Prescribed Forms

None

Adopted Forms

AF Form 3616, *Daily Record of Facility Operation*

DD Form 1801, *DoD International Flight Plan*

Abbreviations and Acronyms

AB—Air Base

AC—Asphalt Concrete

AE—Aeromedical Evacuations

ACC—Air Control Center

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFPD—Air Force Policy Directive

AFRIMS—Air Force Records Information Management System

AGL—Above Ground Level

AGOW—Air Ground Operations Wing

AIP—Aeronautical Information Publication

ALAN—Aircraft Landing Authorization Number

ALS—Approach Lighting System

ALSF—Approach Light System with Sequenced Flashing Lights

AM—Airfield Management

AMC—Air Mobility Command

AMCC—Air Mobility Control Center

AMD—Air Mobility Division

AMOW—Air Mobility Operations Wing

AMXS—Aircraft Maintenance Squadron

AOB—Airfield Operations Board

AOC—Air Operations Center

AOF—Airfield Operations Flight

AOR—Area of Responsibility

AP—Area Planning

APS—Aerial Port Squadron

AS—Airlift Squadron

ASR—Airport Surveillance Radar

ATAP—Air Traffic Act Permit

ATC—Air Traffic Control

ATIS—Automatic Terminal Information Service

AW—Airlift Wing

BAK—Barrier Arresting Kit

BASH—Bird/Aircraft Strike Hazard

Bldg.—Building

BWC—Bird Watch Condition

CAT—Category

CCTLR—Chief Controller

CE—Civil Engineering

CEG—Civil Engineering Group

CES—Civil Engineering Squadron

CFR—Code of Federal Regulation

CMA—Controlled Movement Area

CONUS—Continental United States

CP—Command Post

CRO—Combat Running Offload

CS—Communications Squadron

CTOT—Controlled Takeoff Time

DASR—Digital Airport Surveillance Radar

DP—Departure Procedure

DME—Distance Measuring Equipment

DoD—Department of Defense

DSN—Defense Switching Network

DV—Distinguished Visitor

DZ—Drop Zone

ELT—Emergency Locator Transmitter

EOD—Explosive Ordnance Disposal

ETD—Estimated Time of Departure

FAAO—Federal Aviation Administration Order

FACC—Fire Alarm Communication Center

FAF—Final Approach Fix

FLIP—Flight Information Publication

FOD—Foreign Object Damage

GCA—Ground Controlled Approach

HATR—Hazardous Air Traffic Report

HCP—Hot Cargo Pad

HHQ—Higher Headquarters

HIRL—High Intensity Runway Lights

HQ—Headquarters

IAF—Initial Approach Fix

IAW—In Accordance With

IC—Incident Commander

ICAO—International Civil Aviation Organization

IDP—Integrated Defense Plan

IFE—In Flight Emergency

IFR—Instrument Flight Rules

ILS—Instrument Landing System

IMC—Instrument Meteorological Conditions

ISP—Installation Security Plan

KIAS—Knots Indicated Airspeed

KMC—Kaiserslautern Military Community

LAIRCM—Large Aircraft Infrared Countermeasures

LZ—Landing Zone

MAJCOM—Major Command

MDS—Mission Design Series

MHz—Megahertz

MOA—Memorandum of Agreement

MOC—Maintenance Operations Center

MOG—Maximum On Ground

MSG—Mission Support Group

MSL—Mean Sea Level

NGA—National Geospatial-Intelligence Agency

NATO—North Atlantic Treaty Organization

NAVAID—Navigational Aid

NB—Narrow-Body

NEW—Net Explosive Weight

NM—Nautical Mile

NOTAM—Notice to Airman

NVD—Night Vision Device

NVG—Night Vision Goggles

ODO—Opposite Direction Operations

OG—Operations Group

OPLAN—Operations Plan

OPR—Office of Primary Responsibility

OPS—Operations

OSS—Operations Support Squadron

PAPI—Precision Approach Path Indicator

PCAS—Primary Crash Alarm System

PMI—Preventative Maintenance Inspection

PPR—Prior Permission Required

POV—Privately Owned Vehicle

PTD—Pilot to Dispatch

RABI—Ramstein Air Base Instruction

RAWS—Radar, Airfield, and Weather Systems

RCL—Runway Centerline Lights

RCR—Runway Condition Reading

RDS—Air Force Records Disposition Schedule

REIL—Runway End Identifier Lights

RMS TACAN—Ramstein TACAN Identifier

RPA—Remotely-Piloted Aircraft

RSC—Runway Surface Condition

RSRS—Reduced Same Runway Separation

RVR—Runway Visual Range

SFCC—Security Forces Control Center

SFL—Sequenced Flashing Lights

SFS—Security Forces Squadron

SCN—Secondary Crash Net

SID—Standard Instrument Departure

SKE—Station Keeping Equipment

SOP—Standard Operating Procedure

STARS—Standard Terminal Automation Replacement System

TA—Transient Alert

TACAN—Tactical Air Navigation

TACC—Tanker Airlift Control Center

TERPS—Terminal Instrument Procedures

UAS—Unmanned Aerial System

UFC—Unified Facilities Criteria

UHF—Ultra High Frequency

UPS—Uninterruptible Power Supply

USAFE—United States Air Forces in Europe

VFR—Visual Flight Rules

VHF—Very High Frequency

VISCON—Low Visibility Conditions

VMC—Visual Meteorological Conditions

WB—Wide-Body

Terms

ILS critical area—A critical area is a specific ground area near a radiating localizer or glide slope antenna array, which must be protected from aircraft and vehicle parking and the unlimited movement of surface and air traffic, to ensure the continuous integrity of the signal received by the user aircraft.

ILS sensitive area—A defined area extending beyond the critical area where parking and/or movement of vehicles, including aircraft, is controlled in order to avoid unacceptable disturbances to the ILS performance. The sensitive area is usually within the aerodrome boundaries. (Defined in the Zentralrichtlinie Militärisches Flugverkehrsmanagement A2-272/2-2000-14).

Low Visibility Condition—Weather conditions under which CAT II and III approaches and landing operations are conducted" per German regulations. (Defined in the Zentralrichtlinie Militärisches Flugverkehrsmanagement A2-272/2-2000-14).

Attachment 2

AIRFIELD DIAGRAM AND CONTROLLED MOVEMENT AREA

Figure A2.1. Airfield Diagram and Controlled Movement Area (CMA).



Attachment 3

AIRCRAFT ARRESTING BARRIER ENGAGEMENT PROCEDURES.

Figure A3.1. Aircraft Arresting Barrier Engagement Procedures.

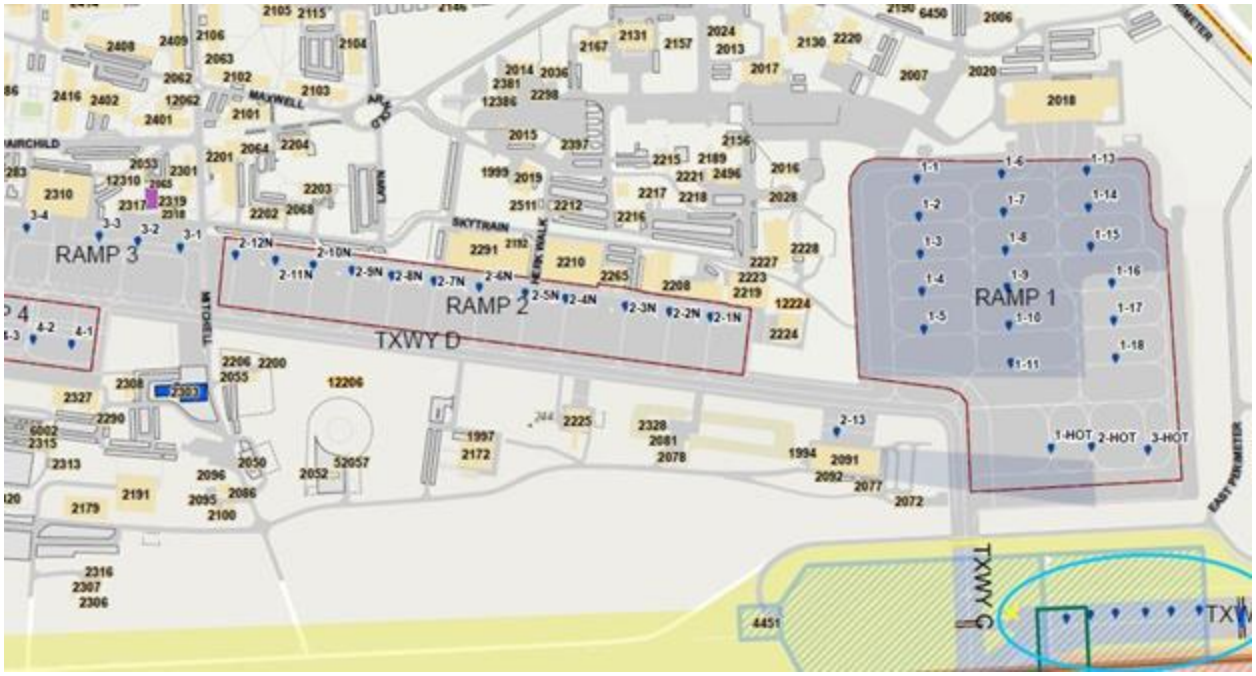
INITIAL RESPONSE AND ASSEMBLY
This will be dictated by the type of barrier engagement.
Fire department vehicles will be positioned at designated locations along the runway according to the established response plan.
Barrier maintenance will report to the applicable aircraft arresting system. After normal duty hours, barrier-qualified Fire Department personnel, if available, will report to the aircraft arresting system.
The following agencies will assemble at the intersection of the road leading past AM and the north crash access road: - Crash recovery. - Ambulance. - AM will position for immediate runway access. - TA will respond for fighter aircraft.
RUNWAY OPERATIONS SUSPENDED
When the aircraft has engaged the barrier, AM will notify tower over the Ramp Net that runway operations are suspended.
AM will notify 86 AW/CP and take appropriate NOTAM action.
RUNWAY RESPONSE
Fire department and crash recovery vehicles will respond to the aircraft upon clearance from the tower.
Barrier maintenance will report to the pits. After normal duty hours and on weekends, qualified Fire Department personnel will report to the barrier pit area.
AM will obtain clearance from the tower and inspect the runway for debris.
All other agencies will respond to the aircraft only at the direction of the incident response commander.
The Fire Department official in command will ensure the aircraft is fire-safe and advise the tower of assistance requirements. The tower will relay the requirement to the AM representative. Vehicles not requested will remain at the assembly point.
The aircraft commander and the 86 OG/CC will determine if the aircraft is capable of self-extraction based on aircraft type and mechanical state.
AIRCRAFT EXTRACTION
Hand signals will be used by fire protection and crash recovery personnel.
Fire department and/or Crash Recovery personnel will remove the aircraft from the barrier cable and contact 786 CES service call for standby response to put the arresting system back in service.
If the aircraft is damaged or disabled, routine priority will normally be used to remove it.
If emergency (30 minutes) or urgent (1+30) priority criteria are to be used, the final decision will be made by the 86 OG/CC.
Note: If prior to the engagement the aircraft commander notifies the tower with intentions to shut down, or determines the situation will warrant shutdown, Fire Department and crash recovery may proceed onto the runway after receiving clearance from the tower.

RESUMING RUNWAY OPERATIONS
As the barrier engagement extraction progresses, the incident response commander will pass estimates to the tower of how long it will take to resume runway operations.
Tower will advise GCA of the anticipated opening time. Tower will advise the incident response commander of any impending emergencies while the engagement clearance is in progress.
As the aircraft is being removed (either by taxi or tow), the AM representative will check the runway for debris and call the sweeper if necessary.
Power Production standby personnel will perform the post-engagement inspection and notify AM/tower when the system is operational. When all vehicles are off the runway, the barrier tape has been rewound, and the runway is clear, the AM representative will notify AM Operations and tower when runway operations can resume. Only Power Production/Barrier Maintenance can return the system to operational status, if they are not available the system will not, under any circumstance, be returned to service. Emergency aircraft will be directed to remaining serviceable arresting system. AM will then notify the 86 AW/CP and take the appropriate NOTAM action.

Attachment 4

AIRFIELD PARKING RAMPS 1, 2, 3

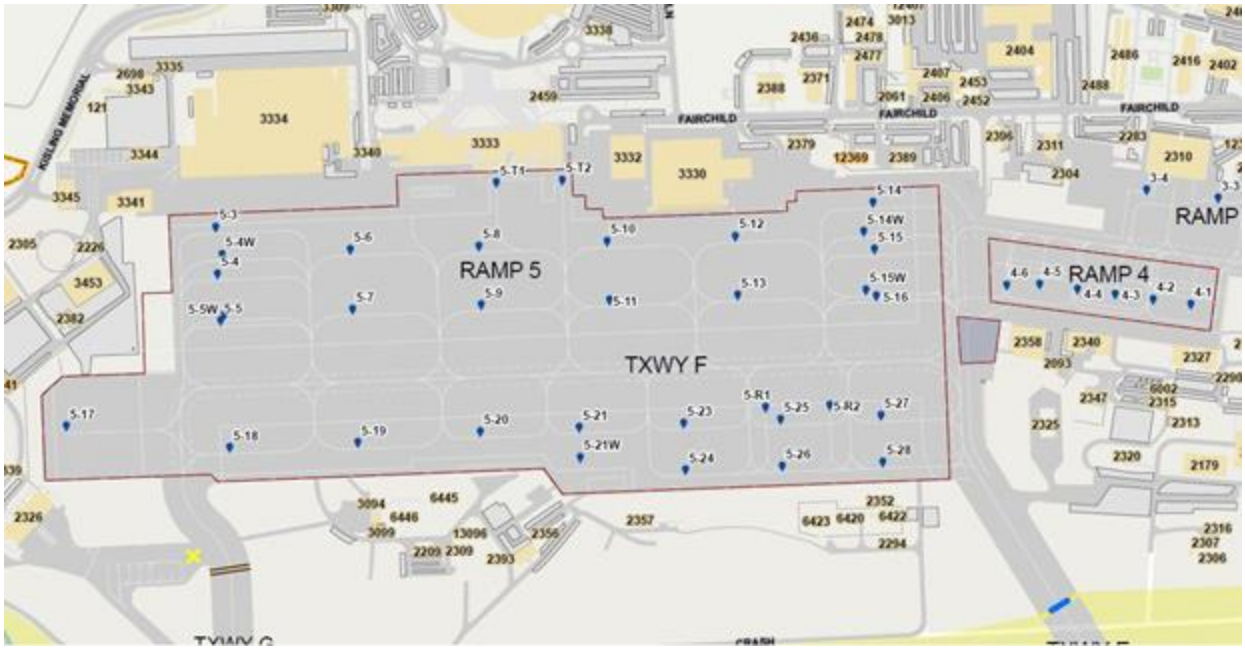
Figure A4.1. Airfield Parking Ramps 1, 2, 3.



Attachment 5

AIRFIELD PARKING RAMPS 4 AND 5

Figure A5.1. Airfield Parking Ramps 4 And 5.



Attachment 6

AIRFIELD PARKING RAMPS 7 AND 8

Figure A6.1. Airfield Parking Ramps 7.

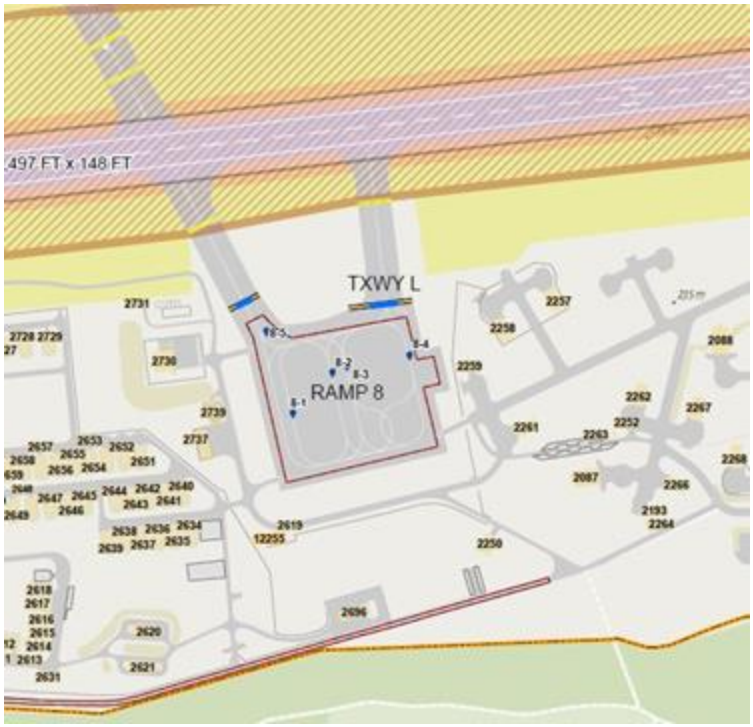
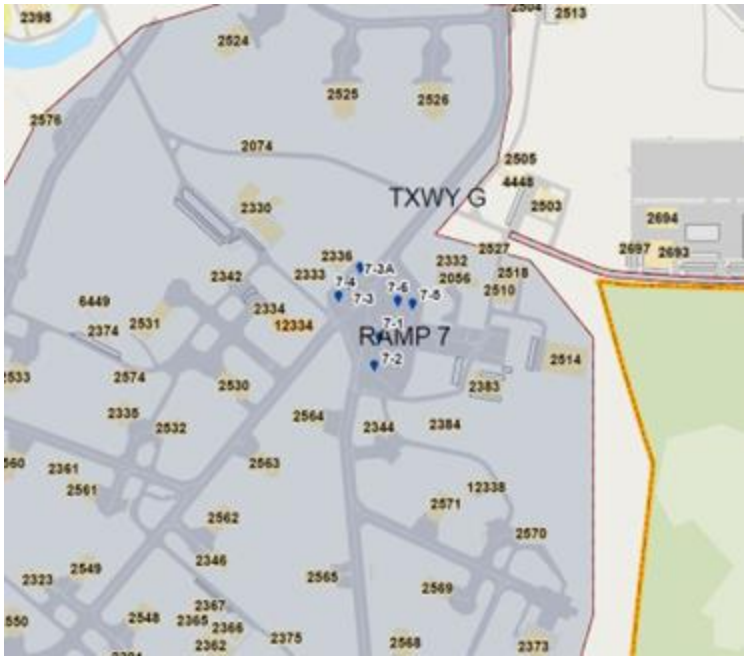


Figure A6.2. Airfield Parking Ramps 8.



Attachment 7

GCA RADAR PATTERN

Figure A7.1. Western Airspace With Extension (Dashed Line).

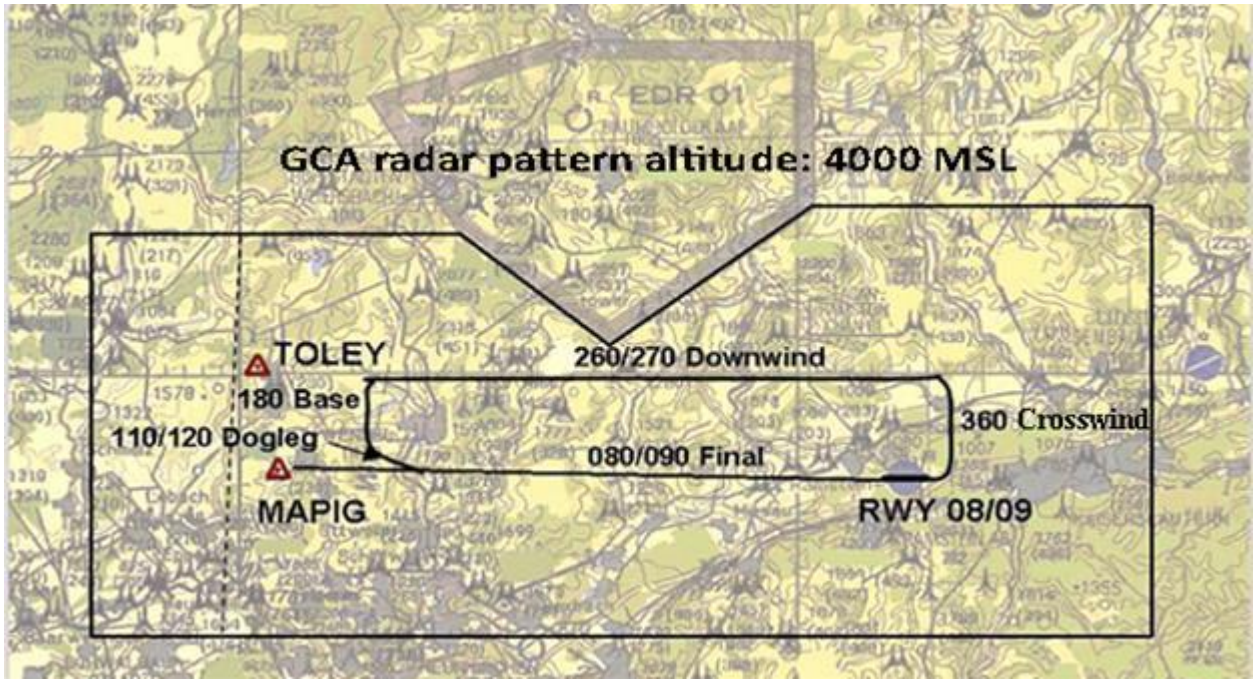
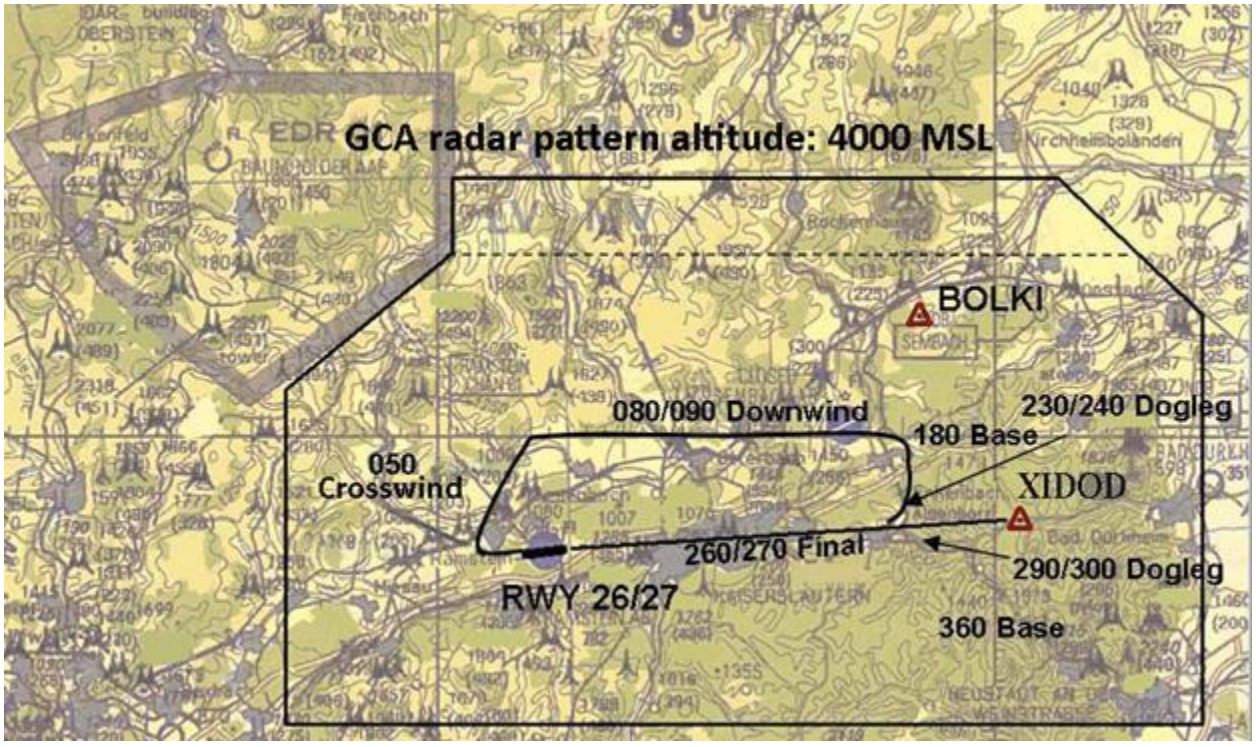


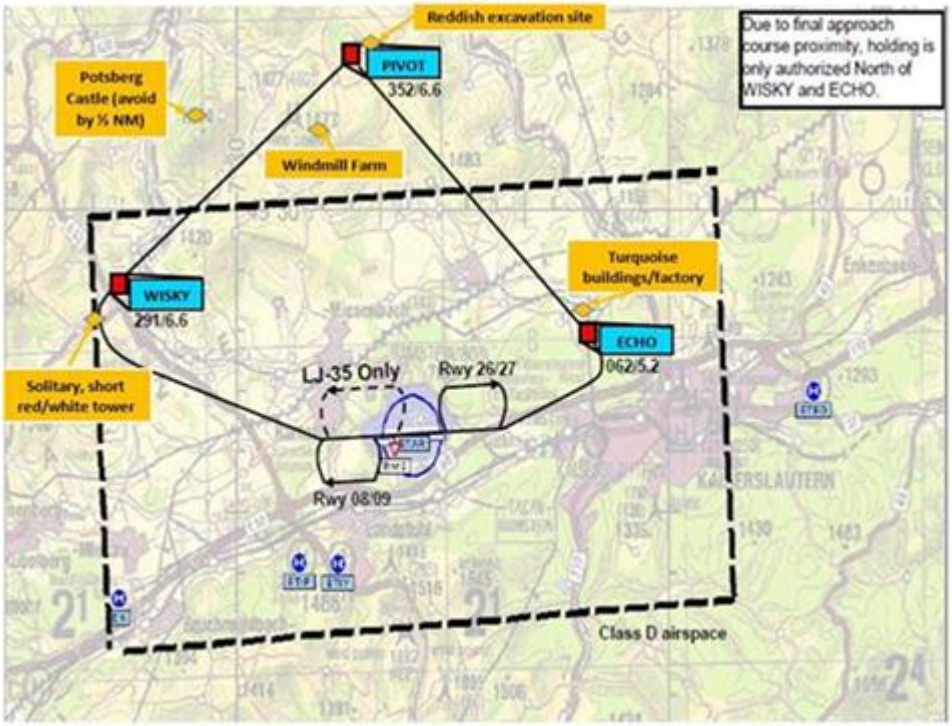
Figure A7.2. Eastern Airspace with Extension (Dashed Line).



Attachment 8

CLASS D AIRSPACE/VFR OVERHEAD/INSIDE/OUTSIDE CLOSED PATTERN

Figure A8.1. Class D Airspace.



Note: ATC will only reference WISKY PIVOT ECHO

Table A8.1. Reporting Points.

REPORT THE FOLLOWING POINTS AT 3000' MSL UNLESS OTHERWISE COORDINATED			
WISKY	AUTOBAHN BRIDGE	RMS 291/6.6	N49-28.47 E07-25.74
PIVOT	GRAVEL PIT	RMS 352/6.6	N49-32.61 E07-33.75
ECHO	ROAD 'X'	RMS 062/5.2	N49-28.83 E07-42.33

Figure A8.2. South Inside/North Inside Closed Pattern.

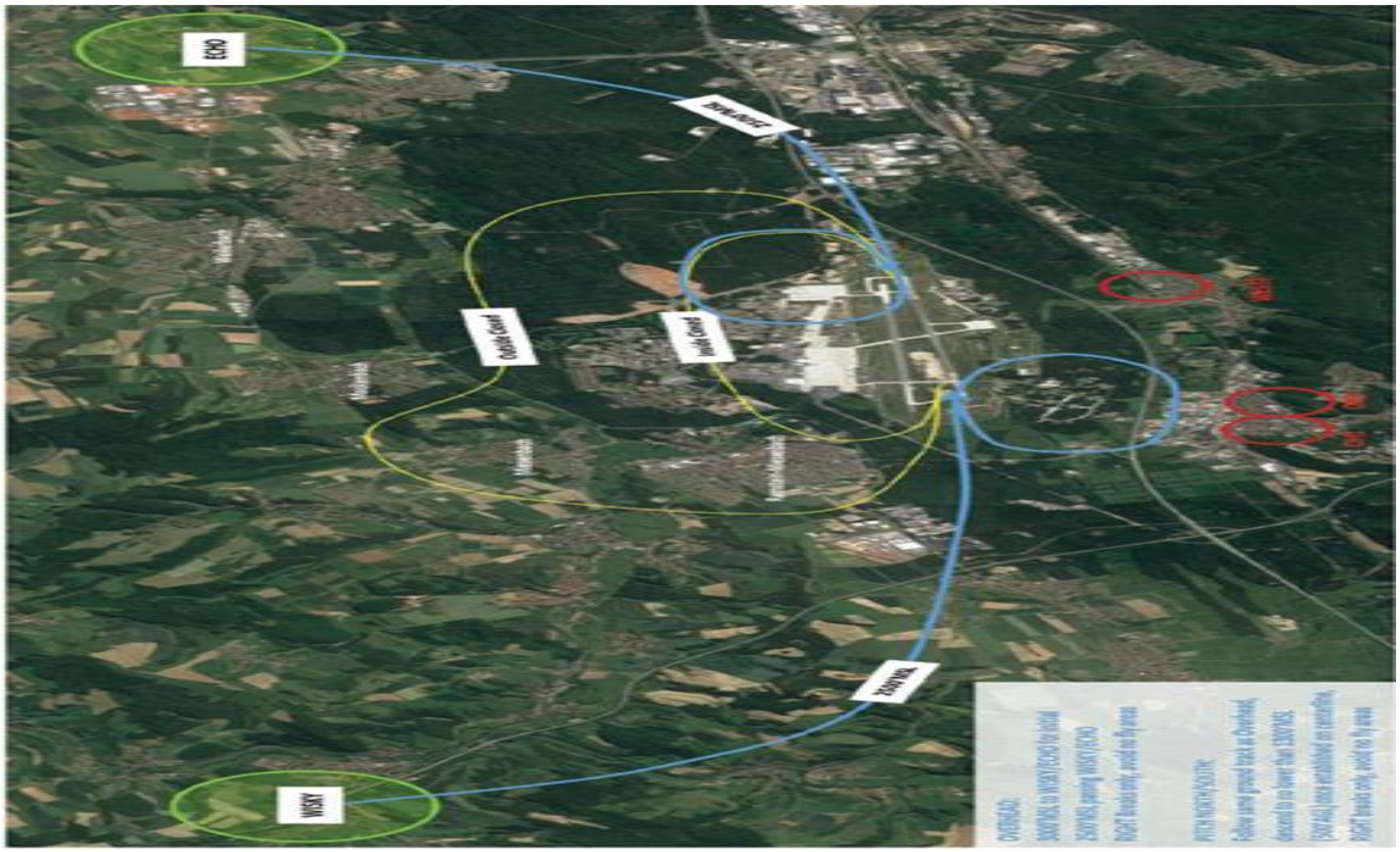


Table A8.2. Populated Area Avoidance.

Vogelweh/Kaiserslautern	Mackenbach	Hβutschenhausen
Otterbach	Misenbach	Spesbach
Erfenbach	Kottweiler-Schwanden	Landstuhl Medical Center
Sigelbach	Glan-Munchweiler	Kindsbach
Rodenbach	Steinwenden	Hohenecken
Weilerbach	Ramstein	

Attachment 9

ILS CAT I/II/IIIA CRITICAL AREAS

Figure A9.1. ILS CAT I Critical Area.



Figure A9.2. ILS CAT II/IIIA Critical/Sensitive Area.



Attachment 10

VFR TACTICAL PROCEDURES (86 AW AIRCRAFT ONLY)

A10.1. VFR Random Steep and Spiral Up/Down Departure.

A10.1.1. Initial - 5,500 feet MSL or coordinated altitude.

A10.1.2. Break - coordinate with tower, 140 KIAS.

A10.1.3. Final - no less than 1/4 mile and 300 feet AGL.

A10.2. VFR Random Shallow Approach.

A10.2.1. May be flown as a straight-in, pitch to the north, or a 90-degree abeam with a 270- degree pitch to final.

A10.2.2. Initial - no less than 1,300 feet MSL (500 feet AGL), 250 KIAS.

A10.2.3. Break to the north at random, 250 KIAS.

A10.2.4. Final - 140 KIAS.

A10.3. Random Shallow Abeam (Runway 26/27 only).

A10.3.1. From PIVOT, proceed south and maintain 3000’ until inside Class D airspace.

A10.3.2. Establish 45-degree entry to intercept gap between Mackenbach and Weilerbach, then descend to 1300 ft MSL (no lower than 500 ft AGL at all times).

A10.3.3. Crews will originate all Runway 26/27 abeam maneuvers no further west than the eastern edge of Ramp 1 (approximately the approach end of RWY 27). Beginning the maneuver from this position avoids inadvertent flight of the WESTF no fly area.

A10.3.4. Caution rapidly rising terrain to the south.

A10.3.5. Not authorized between sunset and sunrise.

A10.3.6. **VFR High Altitude Tactical Straight-In Approach.**

A10.3.6.1. Straight-in, altitude as coordinated, 300 KIAS.

A10.3.6.2. Descend at 10 NM, 3,000 feet per minute descent, 250 KIAS.

A10.3.6.3. Slow descent when reaching 4,300 feet MSL (3,500 AGL).

A10.3.6.4. Final - 1,300 feet MSL - 1,600 feet MSL (500 feet - 800 feet AGL), 3 degree glide path.

A10.4. VFR Low Altitude Tactical Straight-In Approach.

A10.4.1. Straight-in, altitude as coordinate (usually 1,800 feet MSL (1,000 feet AGL)), 290 KIAS.

A10.4.2. Descend at 3 NM, no more than 170 KIAS. A10.4.3. Final – 3 degree glide path.

A10.5. VFR Curvilinear Approach.

A10.5.1. Base slowdown – 9-10 NM, altitude as coordinated (usually 8,800 feet - 10,800 feet MSL (8,000 feet – 10,000 feet AGL)), 200 KIAS.

A10.5.2. Downwind slowdown – 8-9 NM, altitude as coordinated (usually 8,800 feet - 10,800 feet MSL (8,000 feet – 10,000 feet AGL), 200 KIAS.

A10.5.3. Base point – 1.5 NM, altitude 3,300 feet MSL (2,500 feet AGL), 150 KIAS.

A10.5.4. Final – 1.5 NM, 145 KIAS.

Attachment 11

VFR DEPARTURE PATTERN

A11.1. VFR Departure Procedures.

A11.1.1. Takeoff Runway 08/09 South

A11.1.1.1. Climb straight ahead to 1.5 DME.

A11.1.1.2. Turn right to heading 170 degrees and report leaving Class D airspace.

A11.1.2. Takeoff Runway 08/09 North

A11.1.2.1. Climb straight ahead to 3 DME.

A11.1.2.2. Turn left to heading 360 degrees and report leaving Class D airspace.

A11.1.3. Takeoff Runway 26/27 North

A11.1.3.1. Climb straight ahead to 2 DME.

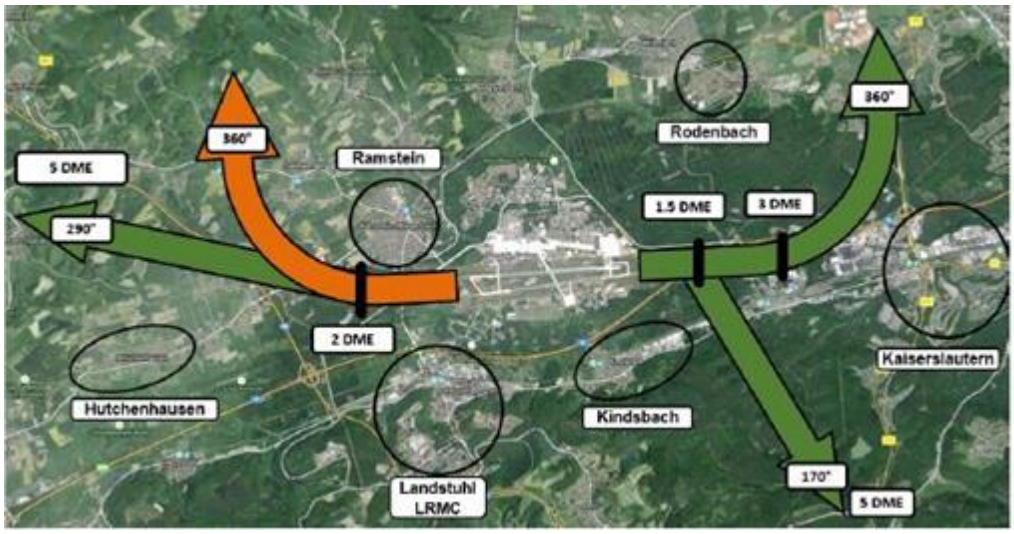
A11.1.3.2. Turn right to heading 360 degrees and report leaving Class D airspace.

A11.1.4. Takeoff Runway 26/27 West

A11.1.4.1. Climb straight ahead to 2 DME.

A11.1.4.2. Turn right to heading 290 degrees and report leaving Class D airspace. **Note:** Fighters must climb to 4000’ MSL or higher for noise abatement purposes.

Figure A11.1. VFR Departure Pattern.



Attachment 12

JETTISON OF EXTERNAL STORES AND CARGO/AIRCRAFT ABANDONMENT AREA

Figure A12.1. Jettison of External Stores and Cargo Runway 26/27.

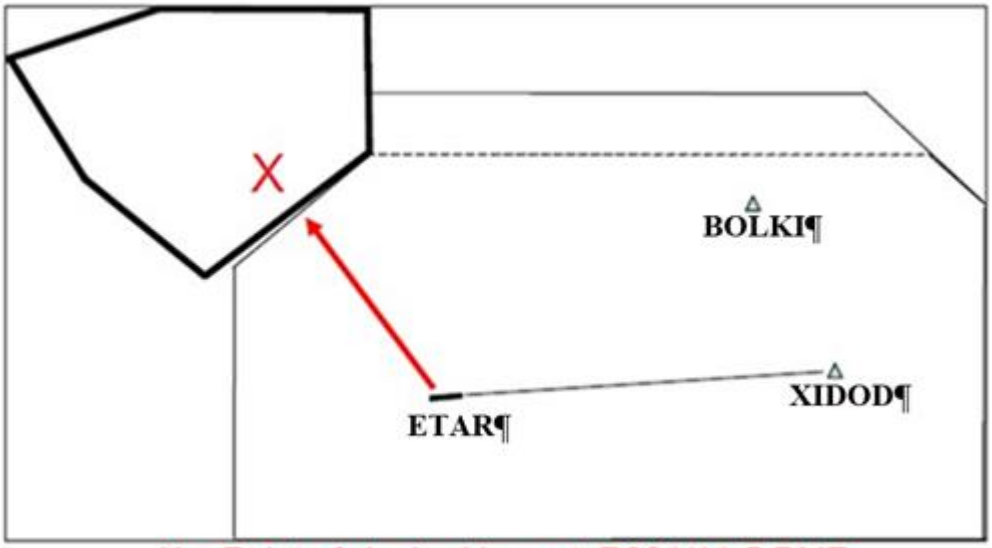
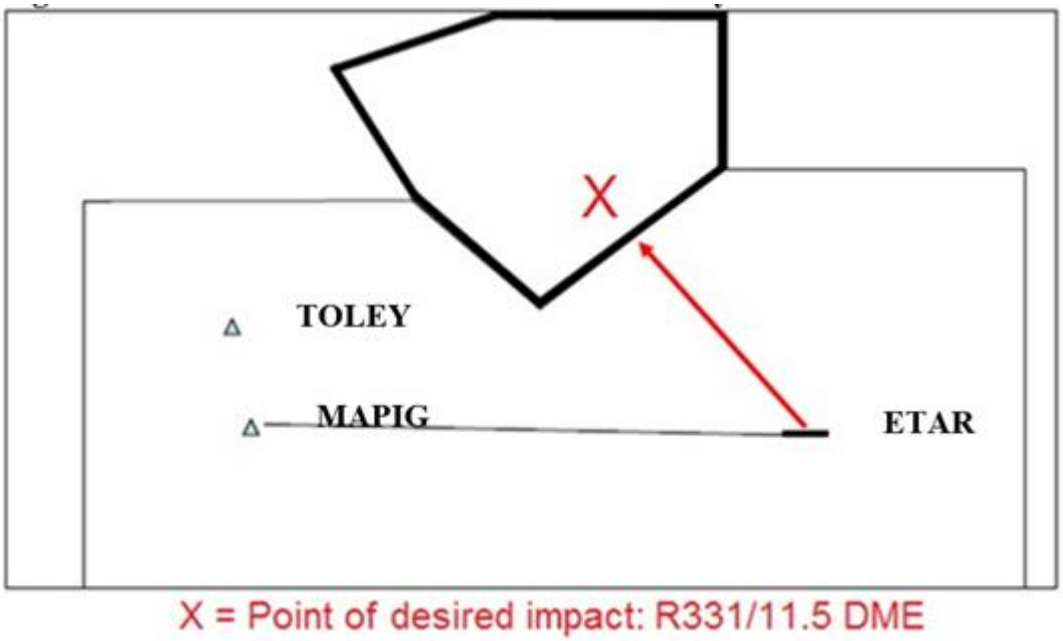


Figure A12.2. Aircraft Abandonment Area Runway 08/09.



Attachment 13

CONSOLIDATED NUMBERS AND FREQUENCIES

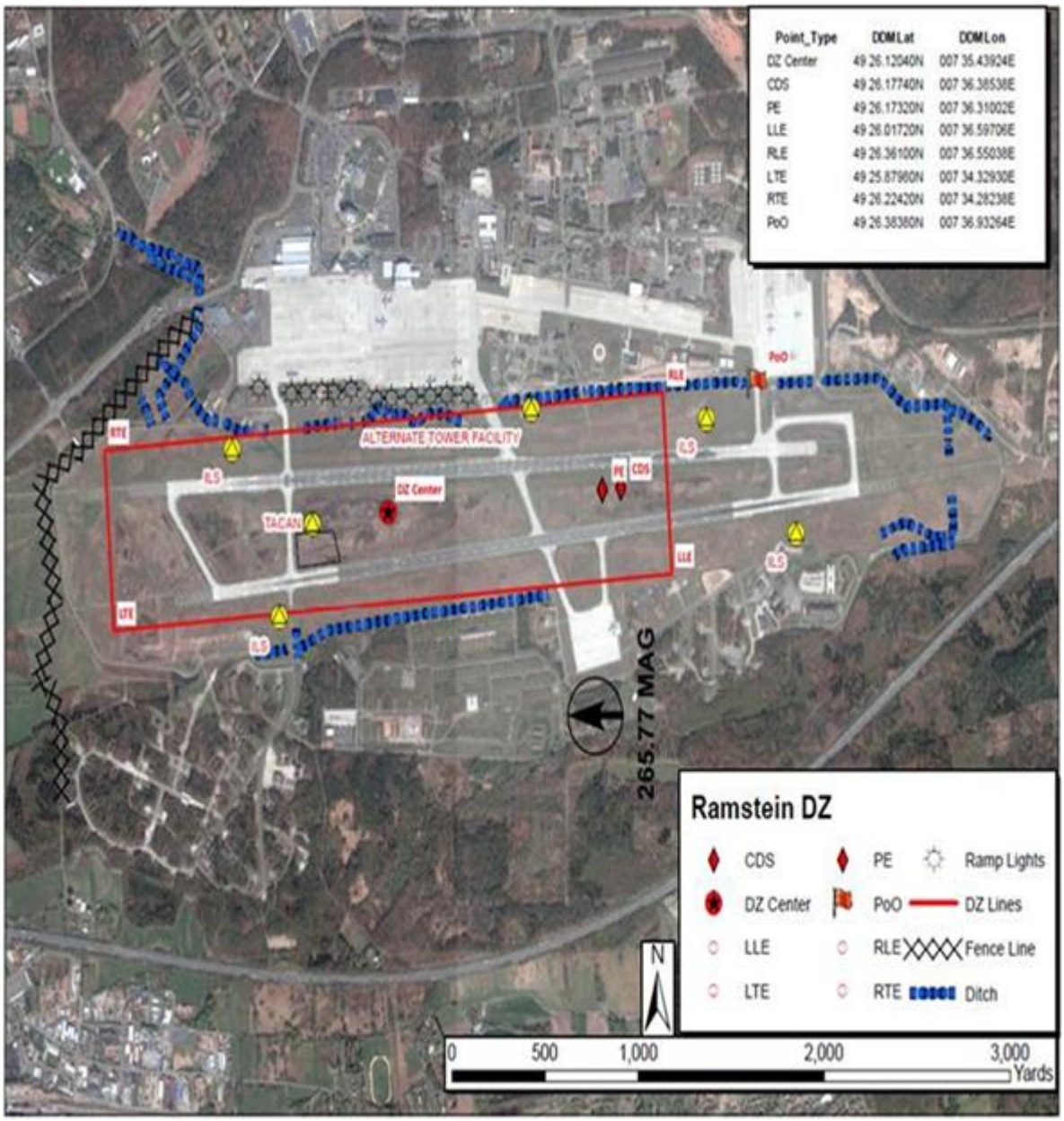
Table A13.1. Consolidated Numbers and Frequencies.

Agency	Number / Frequency
86 Security Forces Pass and Identification	DSN: 480-5429
86 OSS/OSO	DSN: 480-5311
86 Security Forces Physical Security (S5SA)	DSN: 480-4721
ATIS	127.185 or 374.375
ATIS DSN	314-478-3827
86 OSS AM Desk	DSN: 480-2073
Baumholder Range Command	06783-188-2213
Langen ACC	129.675 or 291.0
Ramstein GCA	124.28/140.9 or 284.675
Ramstein Tower	133.205 or 356.85
Ramstein Ground	121.775 or 300.65

Attachment 14

RAMSTEIN AB DROP ZONE DEPICTION

Figure A14.1. Ramstein Drop Zone Depiction.



Attachment 15

QUIET HOUR WAIVER

A15.1. Approval of a Quiet Hour Waiver request DOES NOT constitute approval to land at Ramstein during runway or airfield closures. Requestors must always reference NOTAMS for the latest information.

A15.2. Please input all requested information in the column on the right. Please contact Ramstein Airfield Management at DSN 314-480-2073 or email **86oss.osaaairfieldmanagementoperations@us.af.mil** with any questions.

Figure A15.1. [86 AW Quiet Hour Waiver Request Form.

Date of request (Today's date)	
Date and time request needs to be approved by	
Date departing home station	
Date & time of event	
Event type (arrival, departure, engine run, engine start)	
Type Aircraft (C-130, B703, etc.)	
Tail Number	
Call Sign	
Mission Type (AE, Channel, Training, etc.)	
DV Code (or N/A)	
Mission Priority (1A1, 1B1, etc.)	
Mission number	
Location arriving from (*do not use ICAO*)	
Location departing to (*do not use ICAO*)	
Justification for request. (detailed)	
MSN impact if not approved	
Have all other options been exhausted. What options have been explored?	
POC Info (rank if applicable, name, phone number, & email)	